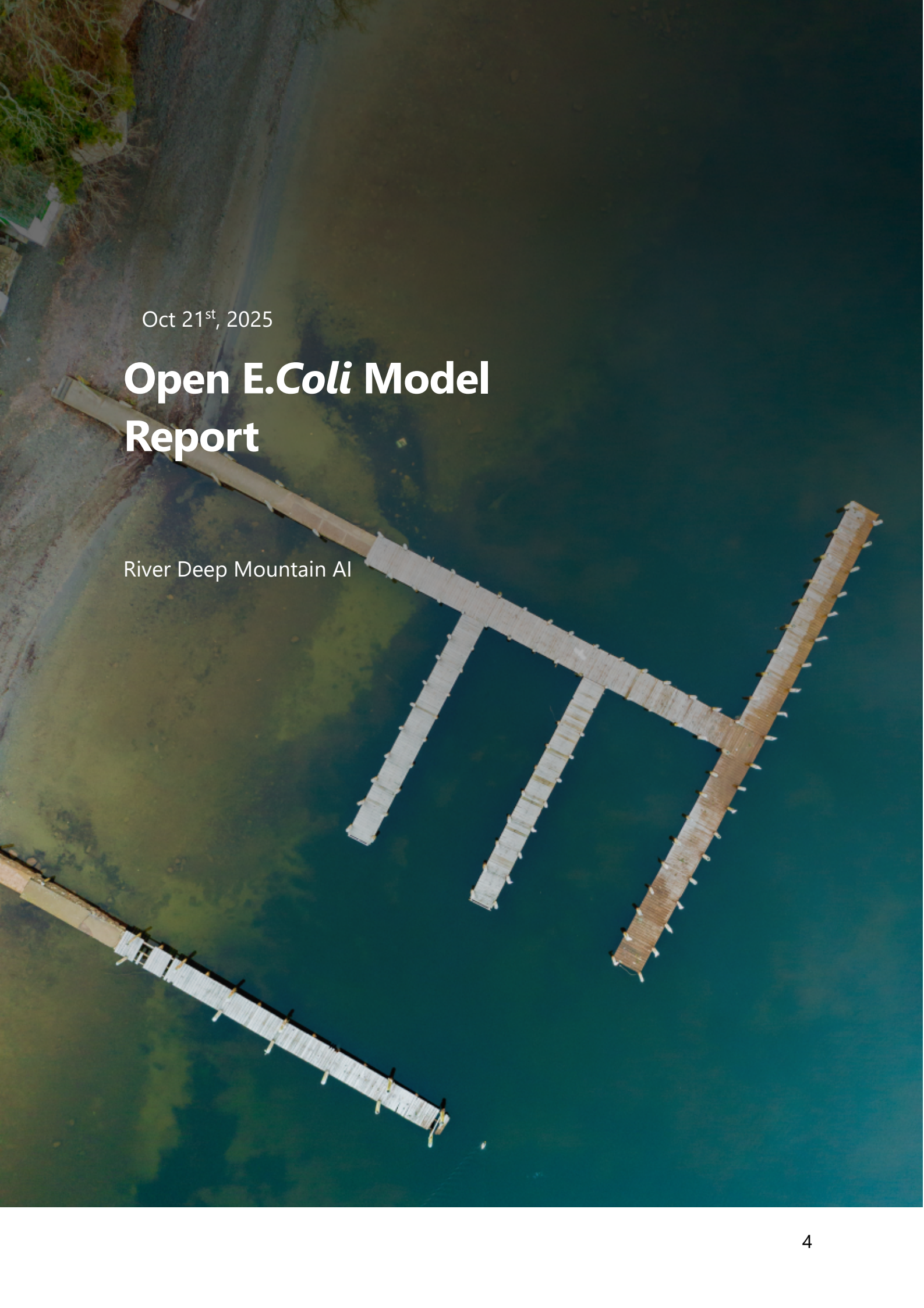


An aerial photograph of a large, multi-armed wooden pier or breakwater structure extending into a body of water. The structure is made of light-colored wooden planks and is anchored with numerous white floats. The water is a deep teal color, and there's a lighter, greenish-brown area near the shore where the pier begins. The sky is not visible.

Oct 21st, 2025

Open *E.Coli* Model Report

River Deep Mountain AI

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1. Introduction

The prediction of *Escherichia coli* (*E. coli*) presence and concentration in recreational bathing waters is important to support public health and environmental monitoring [5]. *E. coli* is a key indicator of faecal contamination and is commonly used to assess the microbial quality of surface waters. While on its own, *E. coli* can pose a risk to human health, its detection also indicates the possible presence of a broader range of faecal pathogens, which may pose additional health risks to humans entering the water [11][12]. Traditional detection methods, while reliable, are often time-consuming and resource intensive. Monitoring *E. coli* is currently constrained by the time it takes to complete a laboratory test.

England currently has more than 450 registered bathing waters that require *E. coli* testing periodically during summer periods to monitor contamination and calculate annual bathing water classifications. The time associated with sampling and analysis often means that the results of tests can take more than 48 hours to be completed and made public. In combination with the periodical sampling and a lack of testing during the winter season, it becomes difficult for swimmers, bathers and other recreational users to decide if the water is safe to come into contact with.

Knowing that bathing water was contaminated by *E. coli* two days ago is not very useful to swimmers or others interacting with a waterbody. If we can provide an accurate forecast of *E. coli*, we can improve people's ability to make an informed decision before entering the water.

To address these challenges, this study explores the application of machine learning (ML) techniques to estimate *E. coli* levels in coastal bathing waters efficiently [1][5]. The objective of this model is to predict *E. coli* concentrations to support bathing water quality assessment. The model should not be used as a lone indicator of human health risk. It is not intended to be a replacement for traditional water quality monitoring, but rather a complementary tool to enhance existing water quality assessment practices.

The models, and their associated performance metrics, presented in this report, represent the first iteration of our ML-models addressing the challenges mentioned above. As River Deep Mountain AI proceeds into phase 3, these models will be developed further and fine-tuned. The results presented here are therefore preliminary and should be considered as such.

1.1 Parallel approach

In this report, we will go through the development and current performance of two different approaches to addressing the challenge mentioned above:

1. ***E. coli* Intensity Estimator:** A ML-model capable of estimating the concentration of *E. coli* in colony-forming units (CFU)/100 ml. This model can later be applied to classify the potential risk of *E. coli* as well and compare performance between the estimator and classifier approach.
2. ***E. coli* Classifier:** An ML-model capable of estimating the risk of *E. coli* contamination based on a pre-defined threshold. In this project, we have tested

models with thresholds of 1000 CFU/100 ml and 500 CFU/100 ml. The threshold of 500 CFU/100 ml is more relevant, as this is the threshold used by the Environment Agency (EA) in coastal bathing waters (see section 3.1).

Furthermore, to make our models to be more widely applicable, we have decided to develop two model versions: an advanced and a light version. This involves training an advanced model with more features and a light version with fewer features, but trained on more *E. coli* samples:

1. **Advanced:** including water quality, land cover, weather and Moderate Resolution Imaging Spectroradiometer (MODIS) features, and trained on bathing water sampling data for locations in England (EA). Accessing *in-situ* water quality features and other environmental features helps the model to achieve high performance in estimating *E. coli*.
2. **Light:** including all features from the advanced version, except water quality features, and trained on bathing water sampling data for locations in England (EA), Wales (NRW), and Scotland (SEPA). Dropping water quality features will make the model more accessible and robust as it does not require any *in-situ* water quality samples.

The following sections will go through the data, development and validation of the advanced and light version of both the estimator and classifier.

1.2 Limitations

Generated classifications or estimation by our model should not be considered as ground truths. Rather, the model should be considered a supportive informative/alerting tool.

- **Imbalance in the data:** Due to the nature of *E. coli* contamination, the data used for training our models are primarily made up of samples containing low or no *E. coli*, with a smaller number of samples of high *E. coli* readings (see Fig. 7 and 12). This creates an inherent imbalance, and risk of bias, in the data and our model, which pulls the model outputs towards lower estimations of *E. coli*. This might impact the models' capability of estimating high concentrations or risk of *E. coli*. Current performances of the models can be seen in section 5. Moving forward, we aim to reduce any imbalance (e.g. adding additional *E. coli* samples).
- **Transferability:** Currently, the models are trained and tested in both temporal and geographical splits, with a notable decline in accuracy when testing the models on unseen geographical areas, which is expected (see section 5). This puts a considerable limitation on the current transferability of the models. As our work continues, we intend to reduce this limitation by introducing

additional *E. coli* samples, adding additional relevant features and re-doing feature selection.

2. Dataset Overview

In our work, we are relying on following key sets of data:

1. Environment Agency (EA) – [Water Quality Archive \(WIMS\)](#).
2. Land cover – [CORINE](#).
3. Weather – [Open Meteo](#).
4. Satellite – [Moderate Resolution Imaging Spectroradiometer \(MODIS\)](#) [17]
5. *E. coli* samples from Natural Resources Wales (NRW)
6. *E. coli* samples from Scottish Environment Protection Agency (SEPA)

The advanced model relies extensively on the first four sources, and the light model combines all six sources, but excludes additional water quality parameters (additional to *E. coli*) to allow the training of the models on additional *E. coli* samples (NRW and SEPA *E. coli* samples; the EA water quality archive is only for England).

In the following sections, we will go through our exploratory data analysis (EDA) for each of the data sources used in our work.

2.1 Environment Agency – Water Quality Archive (WIMS) – Dataset

The EA dataset used for training our models contains **68 million samples** from **2,582 locations** in England, spanning data from the year 2000 to June 2024. Out of these, around **157k samples** specifically contain *E. coli* measurements. However, filtering based on selected proxy parameters (see Table I) reduces this to **1,000–20,000 samples**, depending on the parameters selected. Initial analysis conducted found no strong correlation between *E. coli* and major water quality parameters. and major water quality parameters.

2.1.1 Key Variables and Statistics

The table below summarises some of the parameters in the EA dataset captured along with *E. coli* samples. The parameter labels, have been extracted from the EA Water Quality Archive and represent the “determinand.label” provided by the EA. These parameters are analysed for the *E. coli* estimation and cover various aspects of water quality.

Table I: Variables and Statistics

Parameter	Count	Mean	Std Dev	Min	25%	50%	75%	Max
<i>E. coli</i> C-MF	157696	15845.47	253715.83	0	10	82	901	100000
BOD ATU	22837	74.28	1543.77	0.5	1.6	3	6.23	110000
Ammonia(N)	48386	2.31	78.57	0.002	0.03	0.047	0.145	12000
N Oxidised	45320	4.31	13.78	0.012	1.07	2.7	5.91	1710

Orthophosphate	45666	0.59	6.33	0.001	0.037	0.093	0.22	713
NH3 un-ion	12673	0.0084	0.1052	1.00E-05	0.00039	0.00074	0.00145	5.28
Temp Water	50854	13.27	4.61	0	9.9	13.6	16.6	48
Cond @ 25°C	14836	2317.0	8529.96	0	317	551	830	192794
pH	20396	7.79	0.54	0.609	7.51	7.85	8.1	13.1
O Diss %sat	18965	89.17	24.71	0	81.5	94	100	876
Sld Sus@105°C	45145	45.73	427.91	1	4.33	8.27	18.9	36300
Phosphate	29094	0.26	0.51	0.02	0.06	0.124	0.28	31
Nitrate-N	39193	4.18	4.59	1.00E-05	1.12	2.78	5.93	92
Nitrite-N	41652	0.042	0.27	0.001	0.0054	0.0144	0.0338	29.9
Hardness	1795	201.48	327.85	3.73	63.45	155	280	6400
Alky pH 4.5	10727	141.30	106.48	5	58.5	136	210	3500
COD as O2	5814	456.49	4380.27	3.78	15	38	81	142000
Phosphorus -P	31021	0.31	4.02	0.001	0.0628	0.13	0.302	627
Turbidity NTU	5357	44.40	352.05	1	3	6.7	16.5	16700
Calcium - Ca	1821	66.85	96.35	1	14.7	52.2	94.4	1500
Zinc - as Zn	3731	124.83	944.47	0.995	5	9.76	24.6	38000
Number Birds	18849	4.97	15.78	0	0	0	3	500
No of dogs	18833	1.10	3.20	0	0	0	1	60
Beach Users	75811	26.99	101.16	0	2	7	25	7300
Sewage Debris	76335	0.0067	0.10	0	0	0	0	3
Salinity	1983	23.79	15.98	0.22	0.3	32.5	34.6	333
OrthophsFilt	8696	0.136	0.28	0.001	0.03	0.083	0.15	8.49
SALinsitu	91622	27.93	12.21	0	30.14	33.76	34.67	97.6
Bathers 100m	74418	5.84	18.20	0	0	0	4	1000
Nitrogen - N	29531	5.06	26.33	0.1	2.01	3.78	6.41	4130

IE Conf	1163	5355.2	149992.	1	10	18	189	999999
	24	0	34					9

Note: List of parameters was considerably longer. Here we brought some relevant parameters.

The table provides a statistical summary of various water quality and environmental parameters based on the EA dataset. Parameters, including *E. coli* levels, Biochemical Oxygen Demand (BOD), nutrient concentrations (like ammonia, nitrogen, and phosphates), pH, turbidity, and various other indicators of water and environmental quality, were analysed. For each parameter, Table I shows the count of observations, mean (average value), standard deviation (indicating variability), and values at specific percentiles (25th, 50th, and 75th) to show the data distribution. Minimum and maximum values illustrate the range for each metric, highlighting potential outliers or extreme values. This statistical overview aids in understanding the central tendency, variability, and spread of the data, which is critical for assessing water quality and environmental health indicators across the EA dataset.

2.1.2 Correlation Analysis

Correlation analysis was used to quantify the strength and direction of relationships between the variables proposed for the *E. coli* model. Two methods were considered to provide a more comprehensive understanding of these relationships. **Pearson Correlation** measures the strength of linear relationships between continuous variables, assuming that the changes in one variable correspond proportionally to changes in the other. **Spearman Correlation** assesses monotonic relationships, where variables increase or decrease together without necessarily having a constant rate of change.

Figures 1 and 2 reflect absolute correlations. In Fig.1, the correlation analysis showed that ***E. coli* concentration does not have a strong linear correlation with major water quality parameters** (such as nitrate, orthophosphate, phosphate, ammonia, etc.; a value close to 1 indicates a strong positive correlation, while a value close to -1 indicates a strong negative correlation, and a value of 0 suggests no correlation), although still these parameters were among the highest ranked parameters in terms of the correlation coefficient. This insight informed model design and selection of features to include only those variables that might exhibit some predictive capacity based on trends rather than high linear correlation.

In the Spearman correlation analysis, several parameters show moderate correlations with *E. coli*, ranging between 20% and 50%. These relationships are monotonic, meaning as one variable increases, the other generally does as well, but not at a constant rate. This can be seen in Fig. 2, where certain water quality indicators – like Ammonia (N), water temperature, and Phosphate – exhibit positive, albeit non-linear trends with *E. coli* levels. These moderate correlations help refine the model by incorporating variables that may capture indirect associations with *E. coli*, enhancing its predictive accuracy.

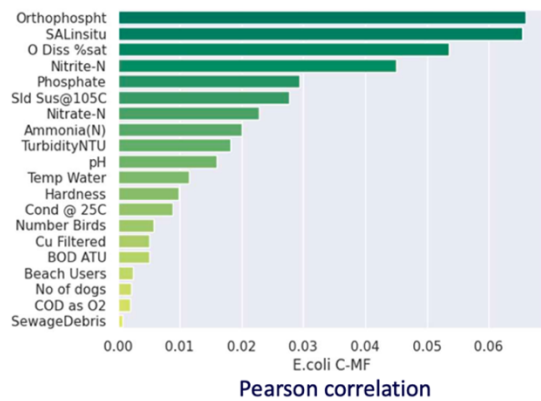


Fig. 1 Displays the Pearson correlation coefficients between *E.coli* and water quality parameters, highlighting the lack of linear relationships in the dataset.

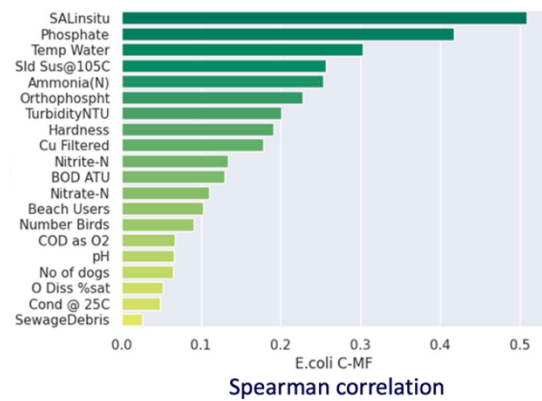


Fig. 2 Displays the Spearman correlation coefficients between *E.coli* and water quality parameters.

One key observation in the Spearman correlation is moderately high relationship between salinity and *E. coli*. Elevated salinity is typically associated with reduced survival of *E. coli* in the environment due to osmotic stress [18].

However, studies have shown that *E. coli* can persist in saline conditions when nutrient availability is high, and high correlations between *E. coli* and salinity have been reported in certain contexts, including saline freshwaters [6,7]. A better understanding of the underlying reasoning behind this correlation (e.g. differences between freshwater and coastal datasets) would be valuable for future development.

Correlations were also observed between *E. coli* and physicochemical parameters such as ammonia and phosphate. Both are closely associated with faecal sources such as sewage effluent, combined sewer overflows (CSOs), and agricultural runoff, which are also key sources of *E. coli* contamination [19].

Laboratory studies have demonstrated that *E. coli* can utilise ammonia as a nitrogen source and that phosphate can influence its behaviour under controlled conditions, though such findings do not necessarily indicate that these nutrients promote significant growth or persistence in natural aquatic environments [8,9,10]. *In situ*, *E. coli* is generally regarded as a transient organism with limited capacity to proliferate outside the host (though exceptions have been reported [20]).

The observed relationships between *E. coli* with ammonia and phosphate are more plausibly attributed to common-source pollution (meaning that *E. coli* originates from the same pollutant sources as ammonia and phosphate) rather than nutrient-enhanced survival or spread, though it should be emphasised that the precise drivers of the observed correlations cannot be conclusively determined.

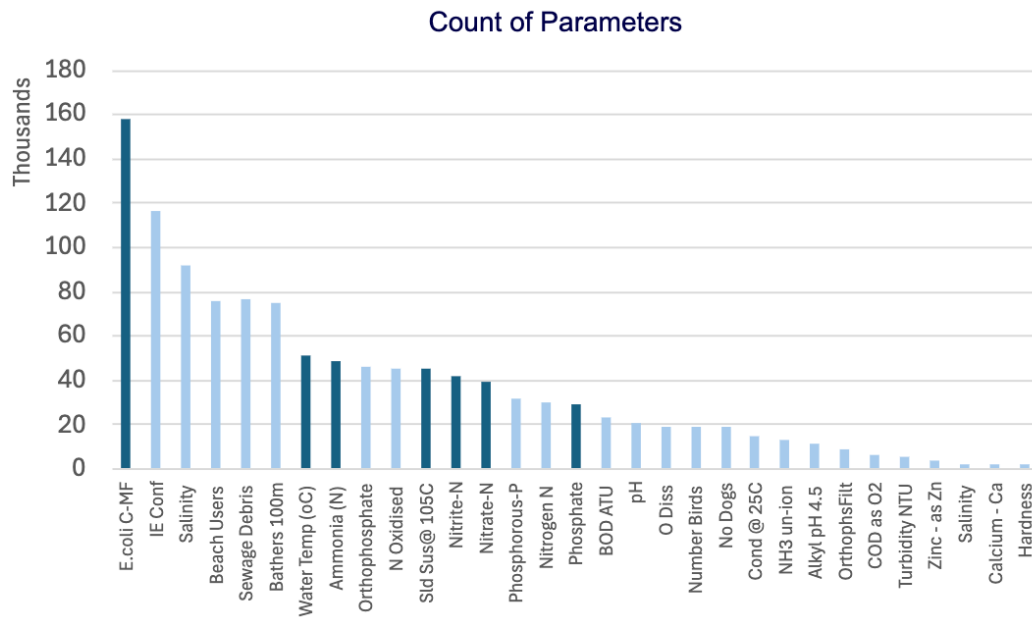


Fig 3. Count of parameters in EA dataset collected at the same day as *E. coli* samples.

Finally, to proceed with our analysis, we selected the 8 parameters for the advanced models that were: (1) correlated (Fig. 2); (2) more frequently reported within the EA dataset (Fig. 3), and; (3) obtainable (but not necessarily available) in near real-time. These were:

1. Water temperature
2. Sld Sus@105°C (Suspended Solids)
3. Ammonia (N)
4. Nitrate-N
5. Nitrite-N
6. Phosphate
7. Daytime
8. Season

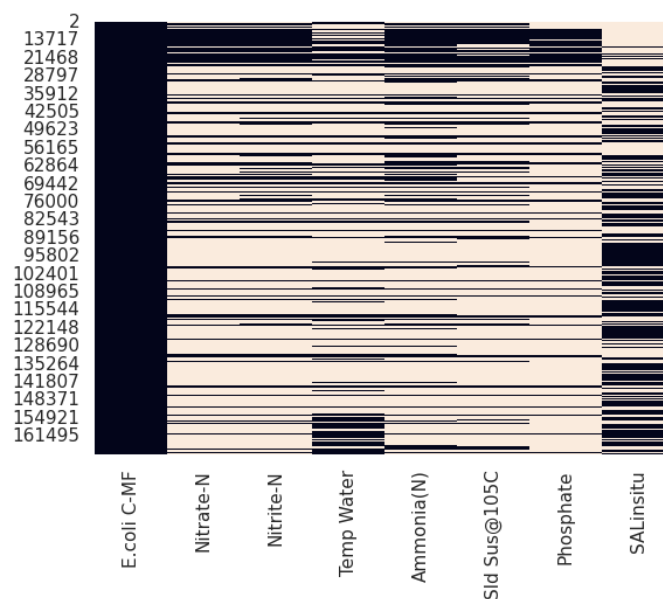


Fig. 4 Presence (dark) and absence (light) of each parameter per sample

While salinity was the most frequently measured parameter, as shown in Fig. 4, when salinity was collected, other key parameters were usually absent. This would imply that analysis considering salinity would be blinded for other key water quality metrics. Therefore, salinity was excluded from our analysis in favour of the remaining 8 parameters chosen. In the end, our filtered dataset contains about **21k samples** from **1,027 locations** in the UK, spanning data from the year 2000 to 2024.

2.2 Land Cover

To enhance the performance of the *E. coli* models, additional datasets were incorporated, including land cover. We have used a land cover dataset covering the European continent that classifies the land surface into 44 classes, ranging from broad forested areas to individual vineyards. The product is called the CORINE (Coordination of Information on the Environment) program and is updated with new status and change layers every six years, with the most recent update made in 2018. This data was used to supplement the water quality features from the EA dataset. We have used the CORINE map with 100-m resolution generated for the year 2018.

Table II CORINE land cover classes; Classes from CORINE have been modified to fit with the 7 classes below¹.

Level 1	Level 2	Level 3
1. ARTIFICIAL SURFACES	1.1 Urban fabric	1.1.1. Continuous urban fabric 1.1.2. Discontinuous urban fabric
	1.2. Industrial, commercial and transport units	1.2. Industrial, commercial and transport units 1.2.1. Industrial or commercial units 1.2.2. Road and rail and associated land 1.2.3. Port areas 1.2.4. Airports
	1.3. Mine, dump and construction sites	1.3.1. Mineral extraction sites 1.3.2. Dump sites 1.3.3. Construction sites
	1.4. Artificial, non-agri. vegetated areas	1.4.1. Green urban areas 1.4.2. Sport and leisure facilities
	2.1. Arable land	2.1.1. Non-irrigated arable land
2. AGRICULTURAL AREAS		

¹ Original classes can be found here: <https://www.eea.europa.eu/data-and-maps/figures/corine-land-cover-1990-by-country/legend>. See more on the modification in the GitHub documents [21].

3. FOREST	2.2. Permanent crops	2.1.2. Permanently irrigated land 2.1.3. Rice fields
		2.2.1. Vineyards 2.2.2. Fruit trees and berry plantations
		2.2.3. Olive groves
	2.3. Pastures	2.3.1. Pastures
	2.4. Heterogeneous agricultural areas	2.4.1. Annual crops (permanent crops) 2.4.2. Cultivation patterns 2.4.3. Land occupied by agriculture, with areas of natural vegetation 2.4.4. Agro-forestry areas
	3.1. Forests	3.1.1. Broad-leaved forest 3.1.2. Coniferous forest 3.1.3. Mixed forest
	4.1. Scrub and/or herbaceous associations	4.1.1. Natural grassland 4.1.2. Moors and heathland
		4.1.3. Sclerophyllous vegetation 4.1.4. Transitional woodland-scrub
5. WETLANDS	5.1. Inland wetlands	5.1.1. Inland marshes 5.1.2. Peat bogs
6. OTHER (Bare lands)	5.2. Marine wetlands	5.2.1. Salt marshes 5.2.2. Salines 5.2.3. Intertidal flats
	6.1. Open spaces with little or no vegetation	6.1.1. Beaches, dunes, sands 6.1.2. Bare rocks 6.1.3. Sparsely vegetated areas 6.1.4. Burnt areas 6.1.5. Glaciers and perpetual snow
7. WATERBODIES (not included)	7.1. Inland waters	7.1.1. Water courses 7.1.2. Water bodies
	7.2. Marine waters	7.2.1. Coastal lagoons 7.2.2. Estuaries 7.2.3. Sea and ocean

To incorporate land cover data into the model, first, for each sampling location, we calculated the percentage coverage of each land cover class within a circular area with a 10 km radius² centered on the sampling location (see Fig. 5). This was done through the following steps:

1. **Buffer Creation:** A circular buffer with a 10 km radius is generated around each sampling point.
2. **Raster Clipping:** The buffer is used to extract the corresponding portion of the CORINE land cover raster.
3. **Class Coverage Calculation:** The number of pixels belonging to each land cover class within the buffer is counted.
4. **Normalisation:** The pixel counts for each class are converted into percentages, representing the proportion of the buffer area covered by each land cover class.

The resulting 44-dimensional feature vector for each location represents the percentage coverage of each land cover class within the 10 km radius.

Then, we generated 7 features from the Land cover feature vector, 6 parental categories (green categories in table above): 1. ARTIFICIAL SURFACES 2. AGRICULTURAL AREAS 3. FOREST 4. SEMINATURAL 5. WETLANDS 6. OTHER (Bare lands), and 7. Dominant Land Cover, the dominant land cover class (i.e. the top class in terms of coverage out of 44 classes). Each category represents percentage of its areal coverage over a radius of 10 km from the sampling location for each sample (Fig. 5).

² The decision of doing a 10 km radius was decided as it provides a faster approach to incorporate and test the impact of land cover data. Further work could beneficially explore the impact of delineating a more precise area of upstream catchments to only include those types of land cover.

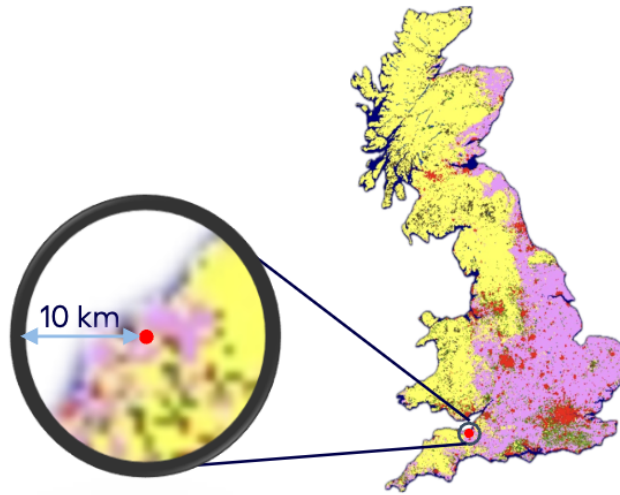


Fig. 5 CORINE land cover map of Britain

2.3 MODIS Data

The MODIS data has been incorporated to provide our models with additional contextual information, to support classification and estimation of *E. coli* risks. This includes Leaf Area Index (LAI), Fraction of Photosynthetically Active Radiation (FPAR), Land Surface Temperature and Emissivity.

MODIS (or Moderate Resolution Imaging Spectroradiometer) is a key instrument to satellites with a 500 m spatial resolution. The MODIS instrument operates on both the Terra and Aqua satellites from NASA. These satellites acquire data in 36 spectral bands. They have a viewing swath width of 2,330 km and cover the entire surface of the Earth every one to two days.

The MOD15A2H Version 6.1 MODIS combined LAI and FPAR product is an 8-day composite dataset with a 500-m pixel size. The algorithm chooses the “best” pixel available from all the acquisitions of the Terra sensor from within the 8-day period [17].

LAI is defined as the one-sided green leaf area per unit ground area in broadleaf canopies and as one-half the total needle surface area per unit ground area in coniferous canopies. FPAR is defined as the fraction of incident photosynthetically active radiation, 400-700 nanometres (nm), absorbed by the green elements of a vegetation canopy.

The Terra MODIS Land Surface Temperature/Emissivity 8-Day (MOD11A2) Version 6.1 product provides an average 8-day per-pixel Land Surface Temperature and Emissivity (LST&E) with a 1-km spatial resolution in a 1,200 by 1,200 km grid. Each pixel value in the MOD11A2 is a simple average of all the corresponding MOD11A1 LST pixels collected within that 8-day period. The 8-day compositing period was chosen because twice that period is the exact ground track repeat period of the Terra and Aqua platforms. Provided along with the daytime and nighttime surface temperature bands are associated quality control assessments, observation times, view zenith angles, and clear-sky coverages along with bands 31 and 32 emissivities from land cover types.

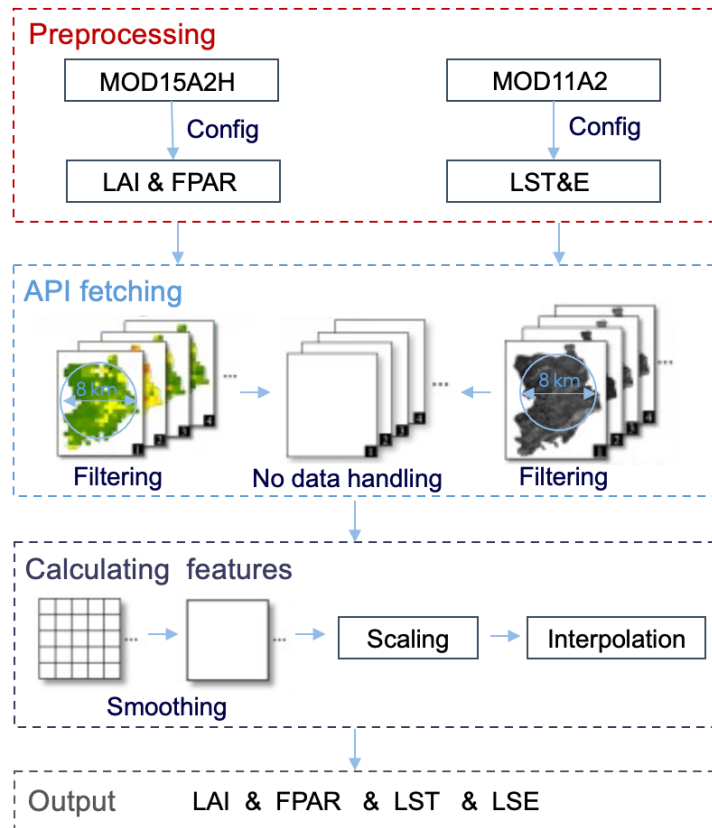


Fig. 6 block diagram of process of extracting MODIS features

With respect to Fig. 6, the data processing workflow begins with preprocessing, where MODIS satellite product (MOD15A2H for LAI and FPAR, and MOD11A2 for LST and LSE) APIs are configured and prepared. The system then proceeds to API fetching, retrieving the datasets while applying filtering to given buffer with a 4km radius (The buffer is used to extract the corresponding portion of the MODIS raster) and handling missing values with searching nearby days if data for the current day was not available. Next, in the feature calculation stage, the data undergoes smoothing (i.e. averaging) to reduce noise, scaling to harmonise values across datasets, and interpolation to fill gaps for continuous coverage. Finally, the processed outputs are generated, consisting of LAI, FPAR, LST, and LSE, which serve as input variables to the models.

2.4 Weather Data

Weather conditions such as temperature, precipitation, and seasonal variations play a critical role in the presence of *E. coli* in water [13]. To capture these effects, historical weather data were obtained using the Open-Meteo Historical Weather API. This resource provides long-term, high-resolution records (approximately 9–11 km grid scale) with up to 80 years of hourly data. Incorporating this data enabled the model to account for the influence of weather patterns on waterborne bacteria levels. Open-Meteo partners with national weather services to bring [14]:

- **High spatial resolution weather data:** Open-Meteo leverages combined global (11 km) and mesoscale (1 km) weather models from established national weather services
- **80 years of historical data:** Accessible via the [Historical Weather API](#) at a 10 km resolution.
- **A free API:** The platform offers free public access for non-commercial use, although the professional subscription version of the API was used in this study to support model training. The free API allows 10'000 API calls per day, 600 per minute.

Open-Meteo compiles information from global and regional weather models produced by national meteorological services, resulting in a large dataset suitable for environmental and epidemiological research. While the platform offers free public access, the professional subscription version of the API was used in this study to support model training.

For each sampling point, defined by its latitude and longitude, weather data was retrieved directly from the Open-Meteo API³. This API utilises global reanalysis datasets (e.g., ERA5), which combine data from satellites, weather stations, and atmospheric models to estimate weather conditions at any given location. This method ensures consistent coverage, even in regions with sparse observational infrastructure. A total of 45 weather-related features were extracted and used as model inputs:

- **Hourly Features (19):** Representing short-term conditions.
- **Daily Aggregates (13):** Summarising daily weather trends.
- **Weekly Aggregates (13):** Capturing longer-term weather patterns.

These features include variables such as temperature, precipitation, humidity, solar radiation, and wind conditions. Full details are listed in Table III (a). These weather data are incorporated, as environmental factors can influence *E. coli* levels. This step involves merging historical weather data with the existing dataset.

While reanalysis-based weather data provides robust and global coverage, it may lack the spatial and temporal specificity of more localised sources, such as high-resolution radar systems. In an ideal scenario, data from weather radars or localised sensor networks could enhance the accuracy and responsiveness of the model to fine-scale weather variations. Similarly, a better understanding and delineation of upstream catchment areas could enhance accuracy and responsiveness of the model. However, due to time and resource constraints, integrating such data sources was not feasible within the current project timeline.

Table III (a): Weather data variables

Variable	Valid time	Unit	Description
temperature	Instant	°C	Air temperature at 2 meters above ground
temperature_max	Daily, weekly	°C	Air temperature at 2 meters above ground
temperature_min	Daily, weekly	°C	Air temperature at 2 meters above ground
relative_humidity	Instant	%	Relative humidity at 2 meters above ground
dew_point	Instant	°C	Dew point temperature at 2 meters above ground
pressures	Instant	hPa	Atmospheric air pressure reduced to mean sea level (msl) or pressure at surface.

³ Similar to the inclusion of land cover, this approach does not consider the variations and size of potential upstream catchment areas. Further work could beneficially explore how to improve this approach.

precipitation	Preceding day and sum (PHDWS)	hour, mm, hours week	Total precipitation (rain, showers, snow) sum of the preceding hour.
rain	PHDWS	mm	Only liquid precipitation of the preceding hour including local showers and rain from large scale systems.
snowfall	PHDWS	cm	Snowfall amount of the preceding hour in centimetres. For the water equivalent in millimetre, divide by 7. E.g. 7 cm snow = 10 mm precipitation water equivalent
cloud_cover	Instant	%	Total cloud cover as an area fraction
shortwave_radiation	PHDWS	W/m ² , MJ/m ²	Shortwave solar radiation as average of the preceding hour. This is equal to the total global horizontal irradiation
direct_radiation	Preceding hour mean	W/m ²	Direct solar radiation as average of the preceding hour on the horizontal plane and the normal plane (perpendicular to the sun)
global_tilted_irradiance	Preceding hour mean	W/m ²	Total radiation received on a tilted pane as average of the preceding hour.
sunshine_duration	PHDWS	Sec	Number of seconds of sunshine of the preceding hour per hour calculated by direct normalised irradiance exceeding 120 W/m ² , following the WMO definition.
Daylight_duration	Daily, weekly	Sec	Number of seconds of daylight per day
wind_speed	Instant, daily weekly (IDW)	& km/h	Wind speed at 10 or 100 meters above ground. Wind speed on 10 meters is the standard level. Max wind speed daily and averaged weekly.
wind_direction	IDW	°	Wind direction at 10 or 100 meters above ground. Dominant direction for daily and weekly.
wind_gusts	IDW	km/h	Gusts at 10 meters above ground of the indicated hour. Wind gusts in CERRA are defined as the maximum wind gusts of the preceding hour. Max wind gusts daily and averaged weekly.
et0_fao_evapotranspiration	PHDWS	mm	ET ₀ Reference Evapotranspiration of a well watered grass field. Based on FAO-56 Penman-Monteith equations ET ₀ is calculated from temperature, wind speed, humidity and solar radiation. ET ₀ is commonly used to estimate the required irrigation for plants.
vapour_pressure_deficit	Instant	kPa	Vapor Pressure Deficit (VPD) in kilopascal (kPa). For high VPD (>1.6), water transpiration of plants increases. For low VPD (<0.4), transpiration decreases
soil_temp_7cm	Instant	°C	Average temperature of different soil levels below ground.
soil_moist_7cm	Instant	m ³ /m ³	Average soil water content as volumetric mixing ratio at 0-7cm, depths.

Link to Open-Meteo Historical Weather API Docs is provided in reference [16].

Figure 7 (below) shows the distribution of data for each of the final variables selected. These variables were considered for development of model v3.0.

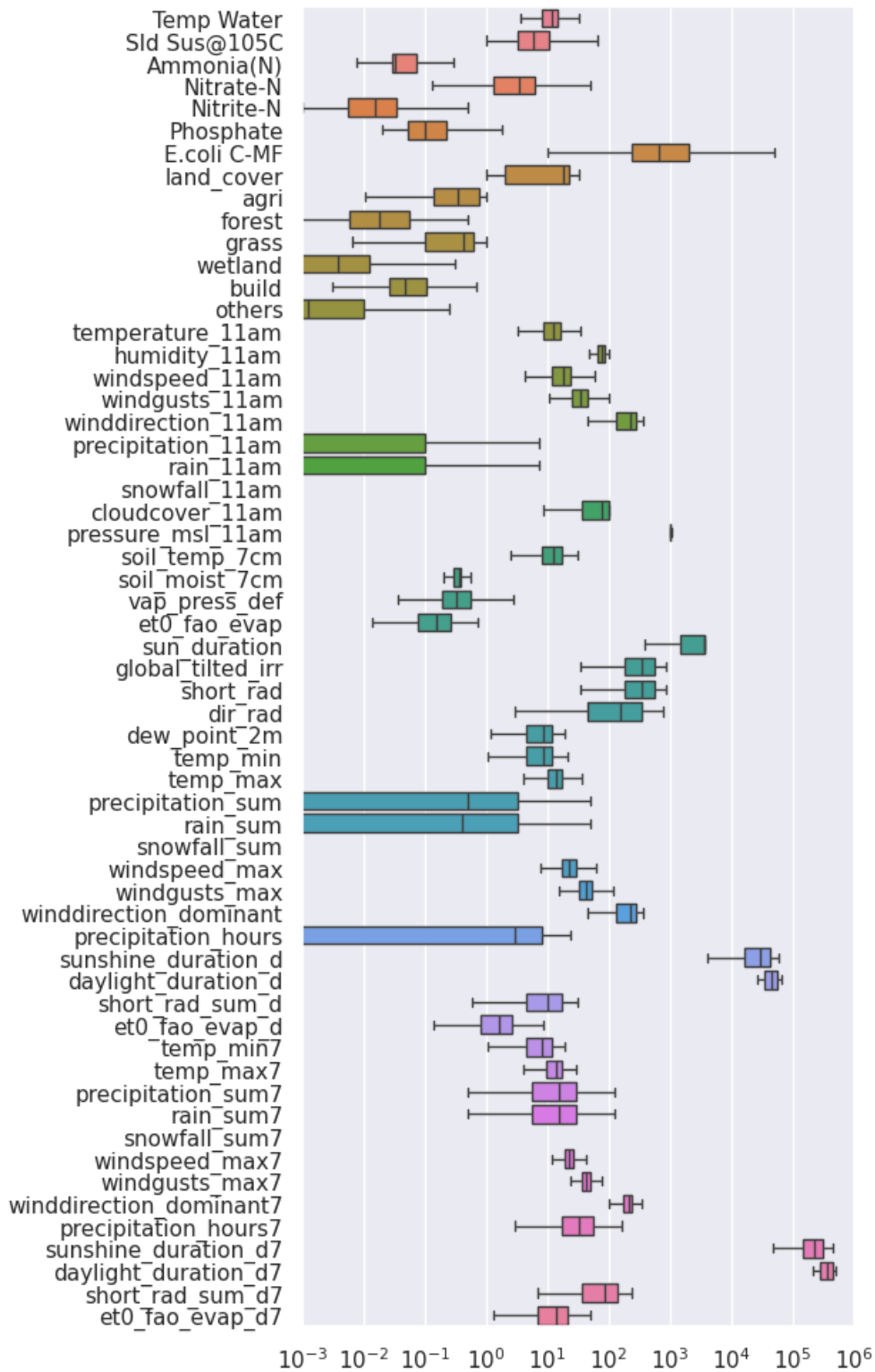


Fig. 7 features distribution box plot

2.5 Additional *E. coli* samples from NRW and SEPA

The SEPA (Scotland) and NRW (Wales) datasets contain approximately 39k and 24k *E. coli* samples, respectively (see Fig. 8). When combined with the *E. coli* measurements from the EA dataset, the total number of *E. coli* measurements reaches around 215k. The light model does not use any of the proxy parameters

from the EA WIMS dataset, which allows the use of all the available *E. coli* samples (from the EA, SEPA and NRW) to train the light model (the EA WIMS dataset only covers England). Below, is a table of all parameters and definition of inputs used to train both models. The light model was trained on the first 59 parameters in Table III (b), while the advanced model used all 65 parameters (i.e. including EA WIMS water quality parameters):

Table III (b) – All parameters used to develop light and advanced version.

	Parameter	Definition	Models
1	Time	Hour of day which the sample was taken	Light/Advanced
2	Landcover	Dominant landcover class – label encoded	Light/Advanced
3	Agri	Percentage of agricultural areas	Light/Advanced
4	Forest	Percentage of forest areas	Light/Advanced
5	Grass	Percentage of seminatural areas	Light/Advanced
6	Wetland	Percentage of wetland areas	Light/Advanced
7	Build	Percentage of artificial surfaces	Light/Advanced
8	Others	Percentage of bare lands	Light/Advanced
9	LST	Land surface temperature	Light/Advanced
10	FPAR	Fraction of photosynthetically active radiation	Light/Advanced
11	EMIS31	Land surface emissivity at band 31	Light/Advanced
12	LAI	Leaf area index	Light/Advanced
13	Temperature_11am	Air temperature at 2 meters above ground at 11 am of sample date	Light/Advanced
14	Humidity_11am	Air Humidity at 11 am of sample date	Light/Advanced
15	Windspeed_11am	Wind speed at 10 or 100 meters above ground at 11 am of sample date	Light/Advanced
16	Windgusts_11am	Gusts at 10 meters above ground at 11 am of sample date	Light/Advanced
17	Winddirection_11am	Wind direction at 10 or 100 meters above ground at 11 am of sample date	Light/Advanced
18	Precipitation_11am	Total precipitation (rain, showers, snow) sum at 11 am of sample date	Light/Advanced
19	Rain_11am	Liquid precipitation at 11 am of sample date	Light/Advanced
20	snowfall_11am	Hourly snowfall sum at 11 am of the sampling date	Light/Advanced
21	cloudcover_11am	Total cloud cover as an area fraction at 11 am of the sampling date	Light/Advanced
22	pressure_msl_11am	Atmospheric air pressure reduced to mean sea level (msl) or pressure at surface at 11 am of the sampling date	Light/Advanced
23	Soil_temp_7cm	Average temperature of different soil levels below ground.	Light/Advanced
24	Soil_moist_7cm	Average soil water content as volumetric mixing ratio at 0-7cm, depths.	Light/Advanced
25	Vap_press_def	Vapor Pressure Deficit	Light/Advanced
26	Et0_fao_evap	ET ₀ Reference Evapotranspiration of a well watered grass field at 11 am of sample date	Light/Advanced
27	Sun_duration	Number of seconds of sunshine Liquid precipitation at 11 am of sample date per hour	Light/Advanced
28	Global_tilted_irr	Total radiation received on a tilted pane as average Liquid precipitation at 11 am of sample date	Light/Advanced

29	Short_rad	Shortwave solar radiation as average at 11 am of sample date	Light/Advanced
30	Dir_rad	Direct solar radiation as average at 11 am of sample date	Light/Advanced
31	Dew_point_2m	Dew point temperature at 2 meters above ground at 11 am of sample date	Light/Advanced
32	Temp_min	Daily minimum temperature at the sampling date	Light/Advanced
33	Temp_max	Daily maximum temperature at the sampling date	Light/Advanced
34	Precipitation_sum	Daily precipitation sum at the sampling date	Light/Advanced
35	Rain_sum	Daily rain sum at the sampling date	Light/Advanced
36	Snowfall_sum	Daily snowfall sum at the sampling date	Light/Advanced
37	Windspeed_max	Daily windspeed max at the sampling date	Light/Advanced
38	Windgusts_max	Daily wind gusts max at the sampling date	Light/Advanced
39	Winddirection_dominant	Daily wind dominant direction at the sampling date	Light/Advanced
40	Precipitation_hours	Daily Precipitation hours sum at the sampling date	Light/Advanced
41	Sunshine_duration_d	Number of seconds of sunshine Liquid precipitation at sample date	Light/Advanced
42	Daylight_duration_d	Number of seconds of daylight per day	Light/Advanced
43	Short_rad_sum_d	Daily shortwave solar radiation as average at sample date	Light/Advanced
44	Et0_fao_evap_d	ET ₀ Reference Evapotranspiration of a well watered grass field at sample date	Light/Advanced
45	Temp_min7	Average minimum weekly temperature	Light/Advanced
46	Temp_max7	Average maximum weekly temperature	Light/Advanced
47	Precipitation_sum7	Weekly precipitation sum at the sampling date	Light/Advanced
48	Rain_sum7	Weekly rain sum at the sampling date	Light/Advanced
49	Snowfall_sum7	Weekly snowfall sum at the sampling date	Light/Advanced
50	Windspeed_max7	Average Weekly windspeed max at the sampling date	Light/Advanced
51	Windgusts_max7	Average Weekly wind gusts max at the sampling date	Light/Advanced
52	Winddirection_dominant7	Average Weekly wind dominant direction at the sampling date	Light/Advanced
53	Precipitation_hours7	Weekly Precipitation hours sum at the sampling date	Light/Advanced
54	Sunshine_duration_d7	Total number of seconds of sunshine Liquid precipitation per week	Light/Advanced
55	Daylight_duration_d7	Total number of seconds of daylight per week	Light/Advanced
56	Short_rad_sum_d7	Weekly sum of daily shortwave solar radiation as average	Light/Advanced
57	Et0_fao_evap_d7	Weekly sum of ET ₀ Reference Evapotranspiration of a well watered grass field	Light/Advanced
58	Distance_to_coast	Distance to the closest coast	Light/Advanced
59	Season	Season of the sample	Light/Advanced
60	Temp Water	Water temperature	Advanced
61	Sld Sus@105C	Solids, Suspended at 105 Celsius degrees	Advanced
62	Ammonia(N)	Ammoniacal Nitrogen as N	Advanced
63	Nitrate-N	Nitrate as N	Advanced
64	Nitrite-N	Nitrite as N	Advanced

65	Phosphate	Phosphate :- (TIP)	Advanced
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Fig 8. Catchment location for (a) NRW and (b) SEPA

3. Strategies

As mentioned in the introduction, we are exploring two approaches to estimating *E. coli* in coastal bathing waters:

1. ***E. coli* Intensity Estimator:** A ML-model capable of estimating the concentration of *E. coli* in CFU/100 ml.
2. ***E. coli* Classifier:** A ML-model capable of estimating the risk of *E. coli* contamination based on a pre-defined threshold. In this project, we have tested models with thresholds of 1000 CFU/100 ml and 500 CFU/100 ml. The threshold of 500 CFU/100 ml is more relevant, as this is the threshold used by the Environment Agency for communicating microbial human health risk in coastal bathing waters (see section 3.1).

3.1 Guidelines for *E. coli* Intensity in Recreational Waters

The *E. coli* classifier in our work is tested using thresholds of 500 CFU/100 ml and 1000 CFU/100 ml to identify high-risk levels of *E. coli* in recreational waters, aligning with international guidelines from the UK, US Environmental Protection Agency (EPA), and the World Health Organisation (WHO).

Table IV: selected regulatory guidelines for *E. coli*

Organization	<i>E. coli</i> threshold (CFU/100 ml) for classification	
UK EA (inland) [2]	≤ 500 (Excellent); ≤ 1000 (Good)	> 1000 (Poor)
UK EA [2] (coastal)	≤ 500 (good)	> 500 (Poor)

US EPA [3]	< 235 (Safe)	< 575 (Partial body contact max)
WHO EU [4]	< 500 (Good)	500-1000 (Moderate); >1000 (High Risk)

The Environment Agency takes up to twenty water samples at each of England's designated bathing waters during the bathing water season between May and September each year. A classification for each bathing water is calculated annually based on samples from the previous four years. For a detailed description on classification thresholds on Inland and Coastal bathing waters in England, see references [2,15].

There are several regulatory benchmarks and guidelines for *E. coli* thresholds in bathing waters globally [2, 3, 4]. They often vary between inland and coastal bathing waters. We have considered guidelines outlined above from the Environment Agency (UK), the Environmental Protection Agency (US) and the World Health Organisation (WHO). Early in the project, we used 1000 CFU/100 ml as a threshold between high and low risk of *E. coli* contamination, but as our work has progressed, we have decided to adopt the coastal-specific guidelines from the EA. This means that for the latest versions of our models, we have also used 500 CFU/100 ml as the threshold for our classifications.

3.2 Data Split & Preprocessing Steps

The dataset is split into training and testing sets, ensuring that the model generalises well to unseen data. In the following, we will distinguish between the advanced and light model, as each have had significantly different numbers of *E. coli* samples available for training and testing (see Fig. 10).

Two training approaches were considered:

Random Geographic Split Training: Trained on most catchments (i.e. 861 for advanced dataset and 3385 for light dataset) and tested on the rest of the unseen catchments (i.e. 152 for advanced dataset and 598 for light dataset). This method is suitable when new unseen sampling points (coastal or inland) are under study.

Random Temporal Split Training: Trained on 85% of data samples for all sampling points and tested on the remaining 15% of the dataset. This method is suitable when current trained catchments are under study. It is known that *E. coli* concentration in the environment is a function of time. However, because the dataset consisted of sparse, irregularly collected samples from multiple monitoring locations, we considered the possibility of treating them as independent observations. To assess whether this was appropriate, we performed an autocorrelation analysis (accounting for uneven time gaps) to determine if past *E. coli* measurements were strongly predictive of future ones.

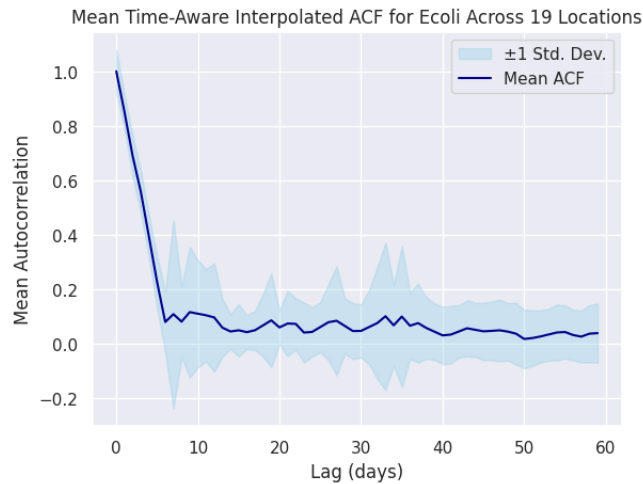


Fig 9 shows the mean time-aware autocorrelation function (ACF) for *E. coli* for 19 locations on EA dataset

The results of the analysis showed (see Fig 9), on average, very weak autocorrelation, suggesting that individual observations can reasonably be treated as independent for modelling purposes. Based on this finding, we proceeded with a random temporal split for model training and testing.

It is important to mention, that only one round of train/test split was done for both the geographical and temporal split. Each split was kept for the entire training, testing and validation phase. This is not expected to have a major impact, as we have so many sampling sites and measurements, however, it does introduce some uncertainty to the performance metrics and validation reported below. Further investigation would be beneficial to estimate the uncertainty of the performance metrics.

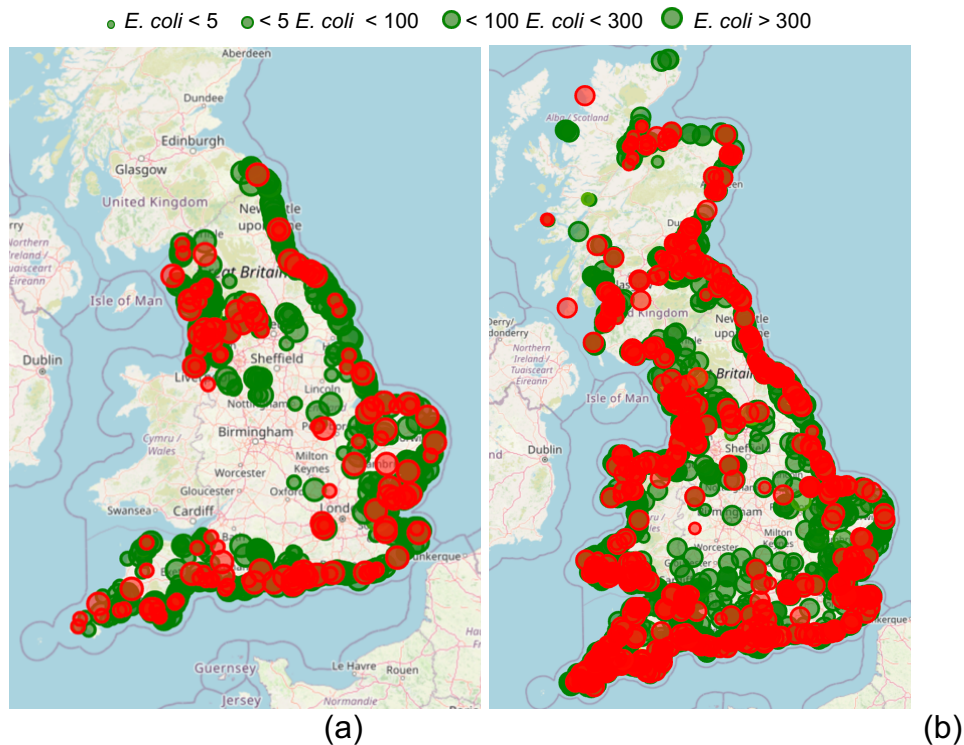
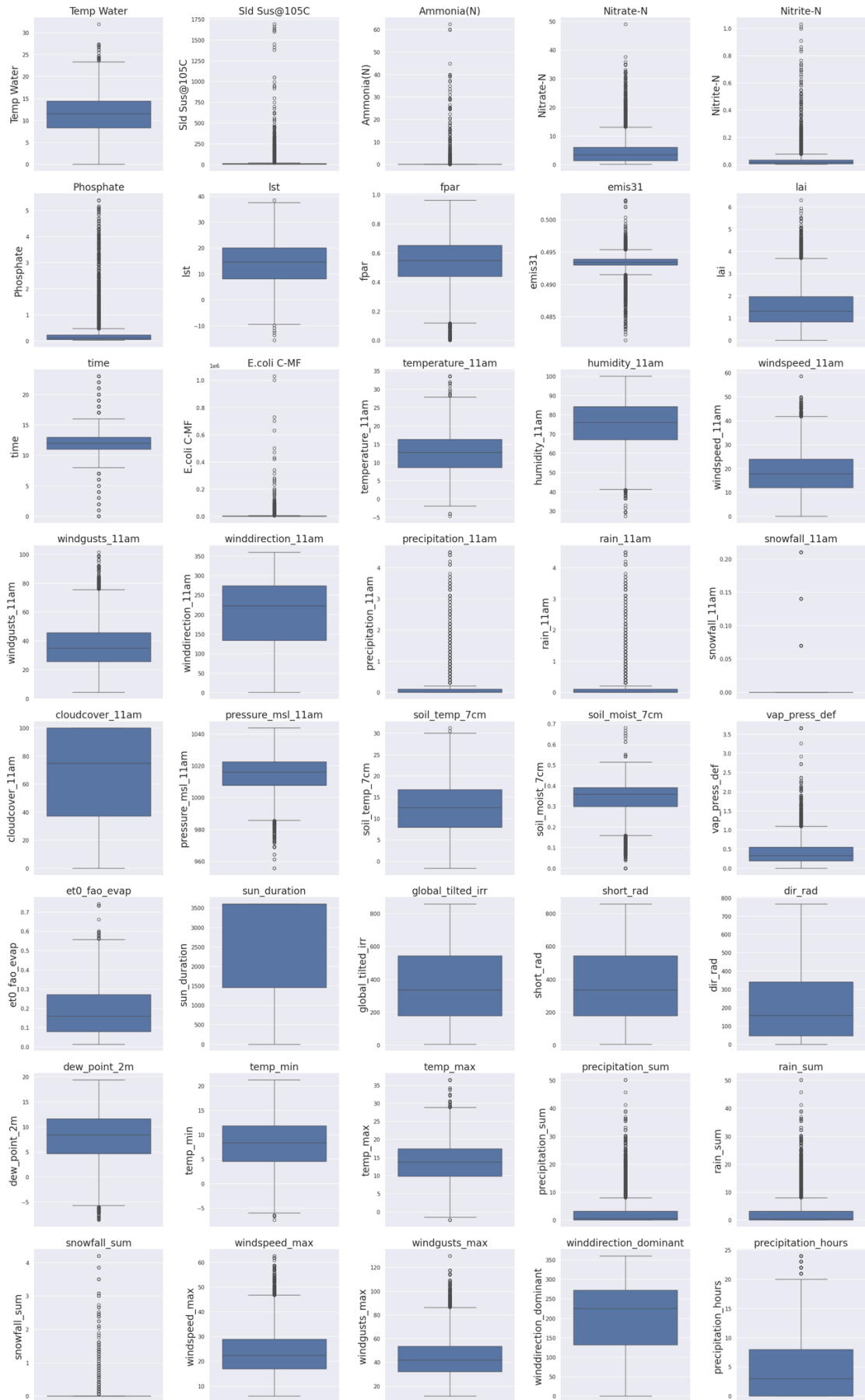


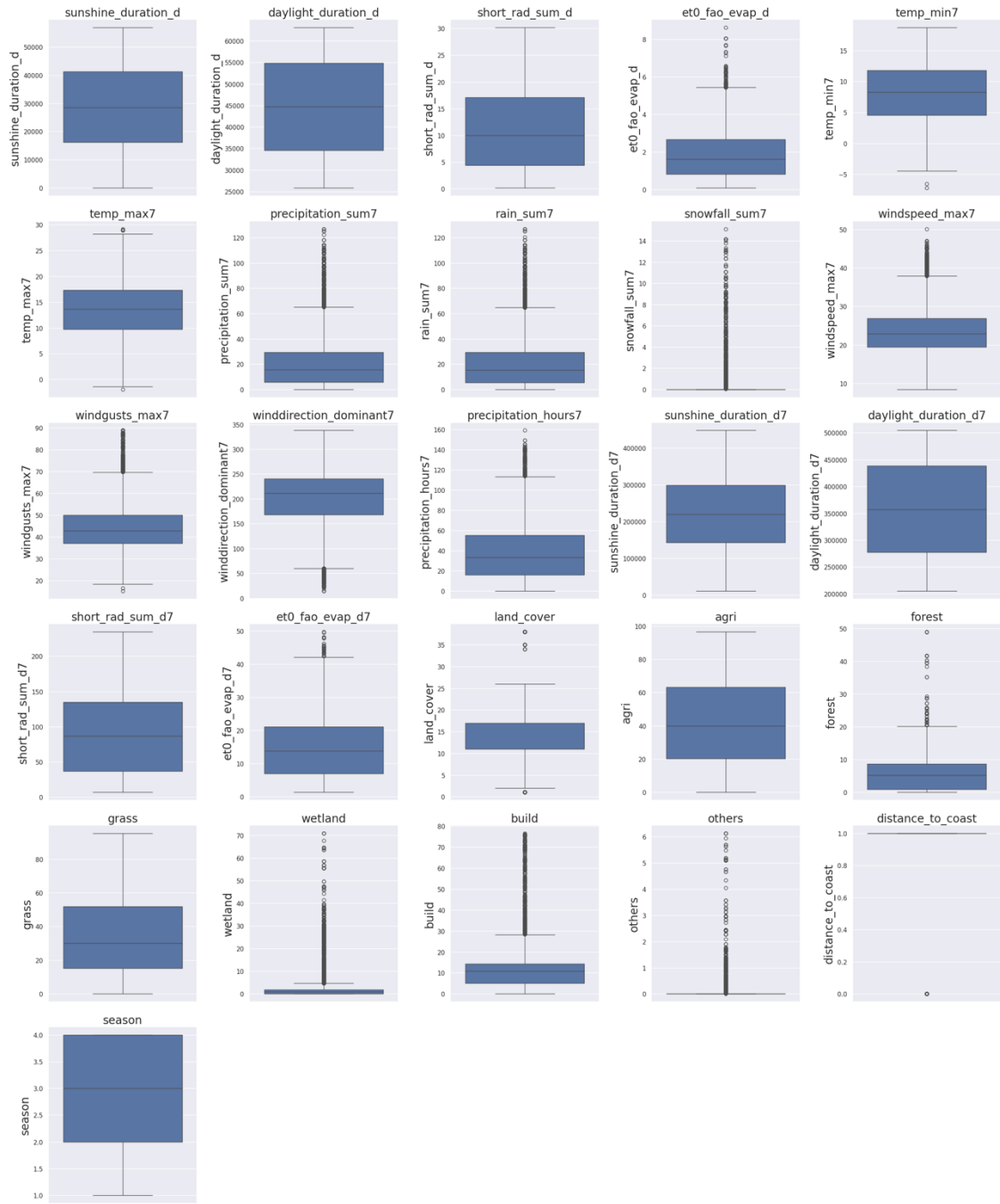
Fig. 10 catchments locations: green dots refer geographic split training locations, and red dots refer geographic split testing locations. (a) advanced dataset (b) light dataset.

Figure 11 (below) displays the distribution of each selected variable using **boxplots** after z-score anomalous data removal.

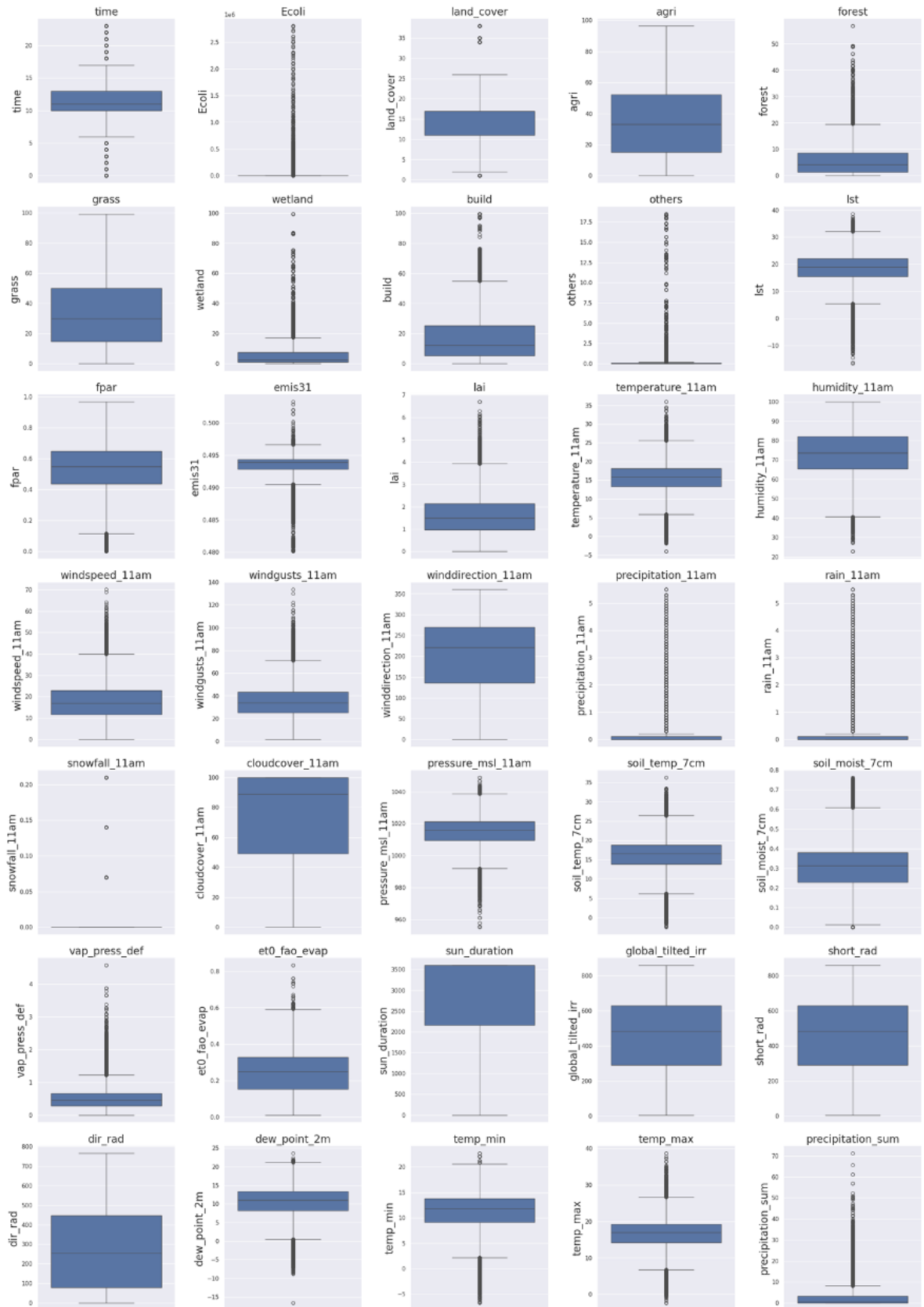
Please refer to Appendix A for the list of parameters and indices that were removed in the outlier removal process. Each boxplot provides a visual summary of the data distribution, focusing on the central tendency and variability:

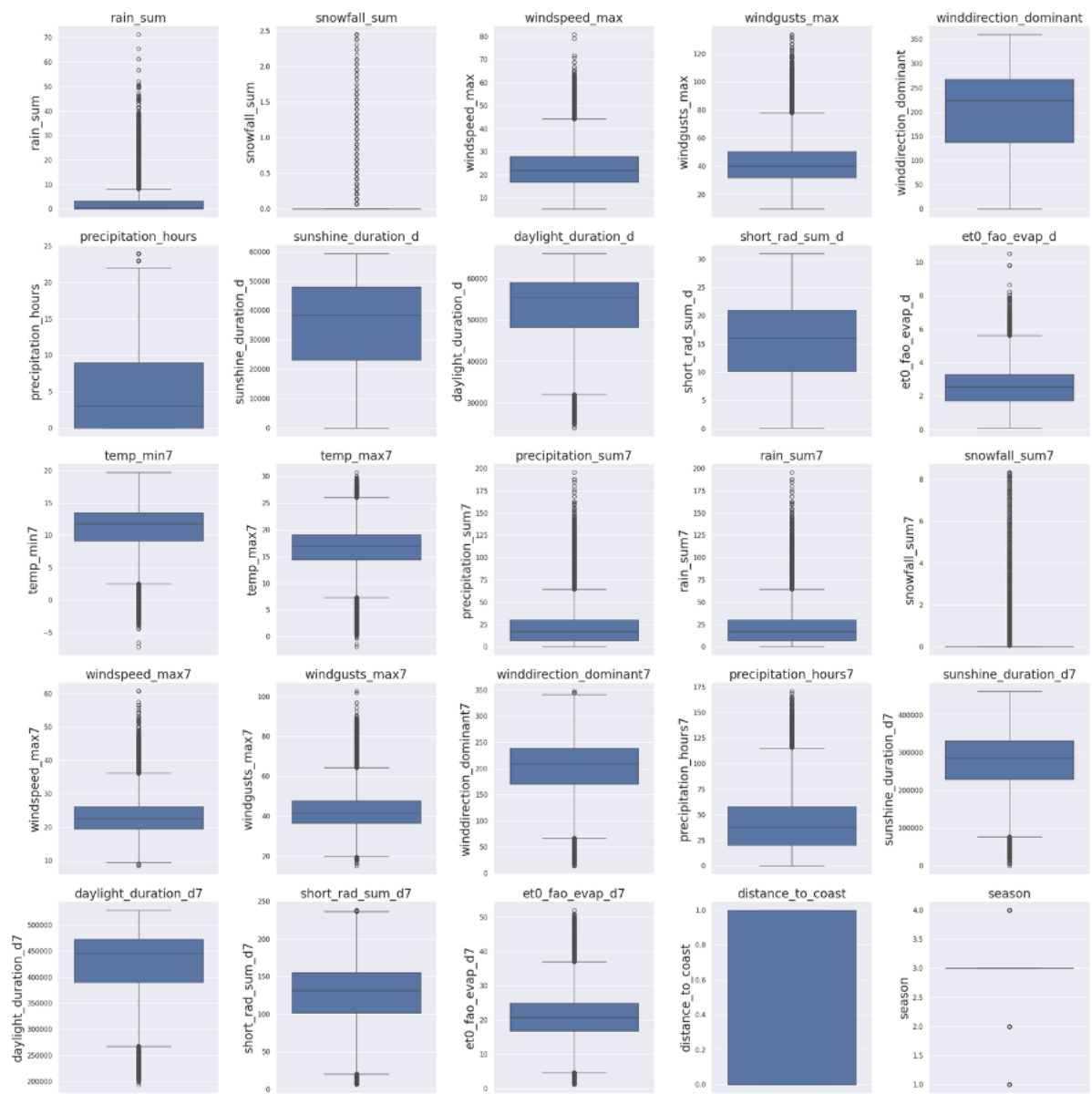
- **Median:** Indicated by a solid black line within each boxplot, representing the middle value, which divides the data into two halves.
- **Box:** Represents the **interquartile range (IQR)** — from the 25th percentile (Q1) to the 75th percentile (Q3).
- **Whiskers:** Extend from the box to show the range of the data within **$1.5 \times \text{IQR}$** from Q1 and Q3. This gives a sense of how far the bulk of the data stretches.
- **Dots:** Shown as black dots outside the box boundaries. Data points that fall **outside the whiskers** (beyond $1.5 \times \text{IQR}$) are plotted individually. These are values that fall beyond the typical range, highlighting data points that are significantly different from the main distribution.





(a)





(b)

Fig. 11 boxplot distribution of the selected parameters. (a) advanced dataset (b) light dataset

To ensure robustness, outliers were dropped using the Z-score method, a statistical approach that removes values lying outside a specified number of standard deviations from the mean. This step refines the dataset by mitigating extreme values that could skew the model and impair its accuracy in predicting *E. coli* levels based on water quality parameters. Any *E. coli* concentration above 10,000 CFU/100 ml is also bounded to this value to make sure the model's focus will remain on the lower concentration levels, i.e. in the range of the EA thresholds (500 CFU/100 ml and 1000 CFU/100 ml).

4. Model Selection

As mentioned above, we are exploring two approaches to estimating *E. coli* in coastal bathing waters:

1. ***E. coli* Intensity Estimator:** A ML-model capable of estimating the concentration of *E. coli* in CFU/100 ml.
2. ***E. coli* classifier:** An ML-model capable of estimating the risk of *E. coli* contamination based on a pre-defined threshold. In this project we have tested models with thresholds of 1000 CFU/100 ml and 500 CFU/100 ml. The threshold of 500 CFU/100 ml is more relevant, as this is the threshold used by the Environment Agency in coastal bathing waters (see section 3.1) and at the same time, from a precautionary perspective, the more conservative threshold is more appropriate for our model.

4.1 Algorithms considered

Algorithms Considered for the Estimator/regression model:

- **XGBRegressor:** XGBoost (Extreme Gradient Boosting) is a powerful ensemble learning algorithm known for its high accuracy and efficiency in handling structured data.
- **LightGBM (Light Gradient Boosting Machine):** is another gradient boosting framework optimised for speed and performance on large datasets.
- **Linear Regression:** A simple statistical model that predicts *E. coli* levels based on a weighted sum of input features. It serves as a baseline comparison for more complex models. It can provide suitable performance when there is strong linear correlation, but it failed in the case of *E. coli* estimation.
- **Support Vector Machines (SVM):** SVM is a supervised learning algorithm that can be adapted for regression tasks. It works by finding a hyperplane in a high-dimensional space that best fits the data while allowing for some error margin. SVMs are effective in capturing non-linear relationships when combined with kernel functions, but they may struggle with very large datasets due to high computational costs.
- **Random Forest:** Random Forest is an ensemble learning method that builds multiple decision trees and averages their predictions to improve accuracy and reduce overfitting. It is robust to noise and can handle complex, non-linear relationships in the data.
- **Multi-Layer Perceptron:** A Multi-Layer Perceptron (MLP) is a type of neural network used for regression. It consists of 3 layers of neurons and can capture complex relationships in the data.

Algorithms Considered for the Classifier model:

- **K-Nearest Neighbors:** The K-Nearest Neighbors (KNN) algorithm classifies data points based on the majority class of their closest neighbours.
- **Support Vector Classification (SVC):** finds the best boundary (hyperplane) that separates different classes. It works well for binary classification tasks.
- **XGBClassifier:** This applies XGBoost for classification, effectively handling complex patterns in *E. coli* presence.

4.2 Evaluation Metrics

- Performance Metrics for the estimator model:
 - Mean absolute error (MAE), R squared (R^2) and root mean squared error (RMSE)
 - In addition to traditional performance metrics for regression models, we have also created a confusion matrix to visualise true vs. predicted classifications with 1000 and/or 500 CFU/100 ml as the classification threshold. We have done this to investigate if the estimation model can be used as a classifier as well.
- Performance Metrics for the classifier model:
 - Accuracy, Precision, Recall, and F1-score,
 - False Negative (FN) and False Positive (FP). Calculation: FN = false negative (i.e. low concentration) samples / number of all samples and FP = false positive (i.e. high concentration) samples / number of all samples. We also provide per class values which is percentage within each class.
 - Confusion Matrix to visualise true vs. predicted classifications with 1000 and/or 500 CFU/100 ml as classification threshold

5. Performance Analytics

In this section, we will go through the current performance of our two approaches: estimation and classification. For each, we will detail the performance metrics for both the advanced and light versions. Initially, we provide an overview of both approaches and their associated performance metrics. In the following sections, we detail the performance of the individual models.

5.1 Overview of performance metrics

In this section, the best model performance information is reported. For full detail on the other algorithms, please see the JupyterLab output pdf file (available on GitHub [21]). Table V (a and b) details the performance for each version of the model (classifier and estimator) from version 1.0 through version 4.0. If applicable, its indicated if the model has been trained with a 500 or 1000 CFU/100 ml threshold and it is indicated if it's an advanced (adv) or light (lgt) version of the models.

The FP and FN columns in Table V show two different values. The primary value represents the percentage of FP and FN relative to all classification outcomes (i.e. considering FN, FP, TN, TP together). The secondary value (in brackets, where relevant) represents percentages within each actual class. In this case, for the positive class, the percentages are calculated based on FN and TP, while for the negative class, they are based on FP and TN.

Table V: The performance of various model versions for *E. coli* classification (table a) and estimation (table b), trained on different subsets of the data.

<i>E. coli</i> Classifier	Random Temporal Split Training						Random Geographic Split Training					
Model _dataset_thr	Accuracy	Precision	Recall	F1- score	FP (Per class)	FN (Per class)	Accuracy	Precision	Recall	F1-score	FP (Per class)	FN (Per class)
Baseline 1.0: XGBC	72.75	0.72	0.73	0.72			70.41	0.68	0.70	0.69		
Baseline 2.0: XGBC	72.78	0.76	0.73	0.70	0.029	0.24	73.33	0.79	0.73	0.70	0.015	0.251
Baseline 3.0: XGBC	78.35	0.81	0.78	0.77	0.023	0.193	73.34	0.77	0.73	0.74	0.054	0.211
XGBCv4 adv 1000	80.77	0.81	0.80	0.80	0.057	0.137	75.29	0.80	0.75	0.76	0.052 (0.17)	0.194 (0.28)
XGBCv4 adv 500	80.86	0.80	0.80	0.80	0.094	0.102	80.36	0.82	0.80	0.81	0.073 (0.40)	0.123 (0.15)
XGBCv4 lgt 1000	87.61	0.87	0.87	0.86	0.030 (0.04)	0.095 (0.44)	80.07	0.77	0.80	0.77	0.049 (0.06)	0.145 (0.70)
XGBCv4 lgt 500	86.74	0.86	0.86	0.86	0.046 (0.06)	0.089 (0.30)	73.81	0.72	0.74	0.73	0.092 (0.14)	0.181 (0.56)

(a) Performance metrics for various versions of *E. coli* classification.

<i>E. coli</i> Estimator	Random Temporal Split Training						Random Geographic Split Training					
Model	Accuracy	MAE	R2	RMSE	FP (Per class)	FN (Per class)	Accuracy	MAE	RMSE	R2	FP (Per class)	FN (Per class)
Baseline 1.0: LGBMR	68.92	1267					64.13	1289				
Baseline 2.0: XGBR	70.66	1244	0.478	2001	0.228	0.065	71.83	1206	1999	0.494	0.189	0.092
Baseline 3.0: LGBMR	73.39	1138	0.556	1859	0.213	0.052	74.89	1595	2399	0.491	0.196	0.054
XGBRv4 adv 1000	81.72	984	0.557	1870	0.090	0.092	77.50	1883	2935	0.282	0.085	0.139
XGBRv4 adv 500	80.51	984	0.557	1870	0.127	0.067	83.49	1883	2935	0.282	0.099 (0.52)	0.056 (0.08)
XGBRv4 lgt 1000	85.61	845	0.497	1808	0.103	0.040	74.13	1227	2349	0.142	0.183 (0.22)	0.095 (0.41)
XGBRv4 lgt 500	82.48	845	0.497	1808	0.145	0.030	65.76	1227	2349	0.142	0.294 (0.39)	0.075 (0.24)

(b) Performance metrics for various versions of *E. coli* estimation.

In the provided table, the highlighted metrics **Accuracy** and **MAE** are emphasised due to their specific relevance in evaluating model performance for *E. coli* estimation and classification. The MAE metric is used in the *E. coli* estimation models to assess the average magnitude of prediction errors, treating all errors with equal importance. Since MAE does not overly penalise larger errors, it provides a balanced view of model performance, focusing on overall consistency in predictions. Large MAE values can be related to high margin errors on larger *E. coli* concentrations.

Accuracy is particularly relevant for our *E. coli* classifier because the task is a binary classification – estimating whether *E. coli* levels exceed a certain threshold. In binary classification, accuracy is straightforward and interpretable, indicating the proportion of correctly classified samples. This metric is effective in this context, especially when the model needs to be reliable in alerting to high-risk *E. coli* levels, thus directly impacting public health advisories. Accuracy provides an easy-to-understand measure of how often the model makes correct predictions.

When evaluating the performance of the *E. coli* classifier it is also important to look beyond overall accuracy. This is because accuracy can be misleading for imbalanced data sets, where high and low risk events occur at different frequencies. In such cases, the model could achieve high accuracy by predicting most sample as low risk and fail to identify true high-risk events. In this case, **precision** and **recall** need also to be considered. Precision measures the number of events that were predicted as high-risk, which were truly high risk, while recall measures how many high-risk events were correctly predicted by the model. In this analysis, we provide weighted average precision and recall, which account for class imbalance by weighting the contribution of each class according to its frequency. This helps provide further insight into the model's reliability in identifying potential health risks.

5.2 Estimator Performance Analysis

In this section, we detail the performance of the XGBRegressor model, capable of estimating *E. coli* concentrations in CFU/100 ml. First, we go through the model performance from a regression point of view. Secondly, we explore the estimation model's capability of classifying the risk of *E. coli*. This allows us to compare it with our classification model, which is detailed in section 5.3.

5.2.1 Regression metrics

To better understand the regression performance of the XGBRegressor model, Figures 12–15 provide complementary perspectives on prediction accuracy across both the advanced and light datasets. Figure 12 shows pseudo time-series comparisons between 100 random samples of observed and predicted *E. coli* concentrations, revealing instances of under- and over-estimation. Figure 13 complements this view with scatter plots of actual versus predicted concentrations, offering a direct assessment of alignment with the 1:1 line; this helps visualise systematic biases and deviations across the prediction range.

To place these results in context, Figure 14 illustrates the distribution of the test data in histogram form, clarifying the underlying frequency of different *E. coli* concentration levels that the model must capture. Building on this, Figure 15

provides a more detailed error analysis: panel (a) presents the distribution of absolute errors, while panels (b) and (c) break down mean and absolute mean residuals across *E. coli* concentration bins. Together, these plots reveal how prediction errors vary across concentration ranges, which is particularly useful for understanding model reliability under low, and high contamination levels. An immediate observation is that at higher concentrations, the predicted values flatten out and relatively model underestimate the true values. This might be that the XGBooster model predictions converge toward the mean of the training data or it might be due to the fact that the dataset has more samples in the mid-range, the model will optimize performance there, not at the tails.

Figure 12 (below) represents the model performance over 100 random samples for the **XGBRegressor** model.

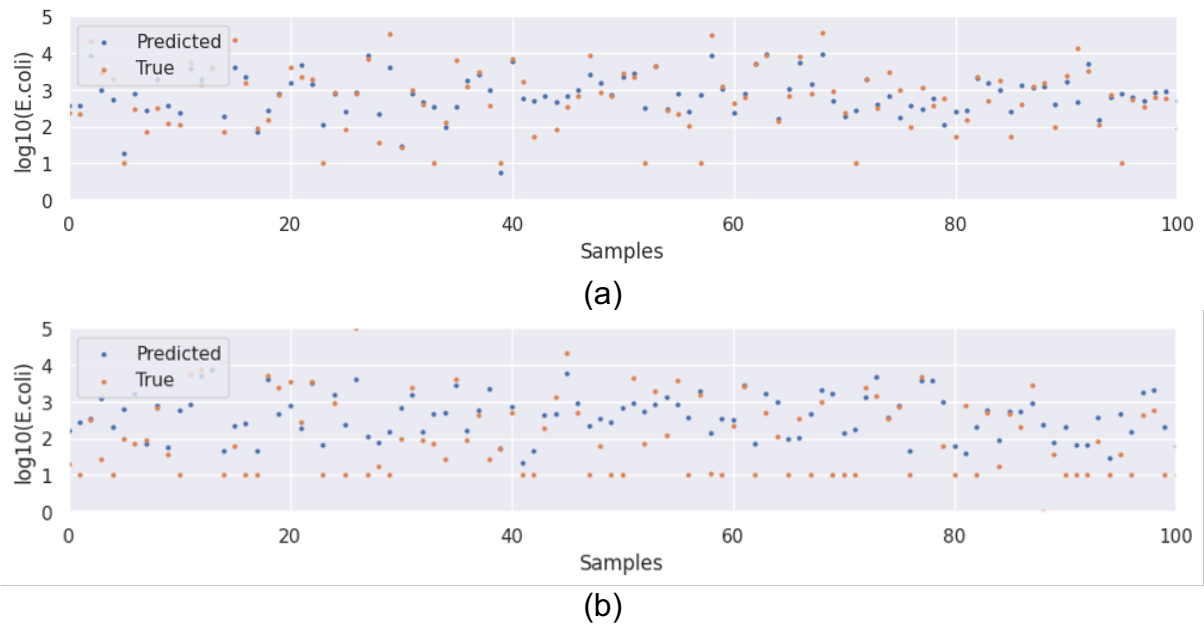
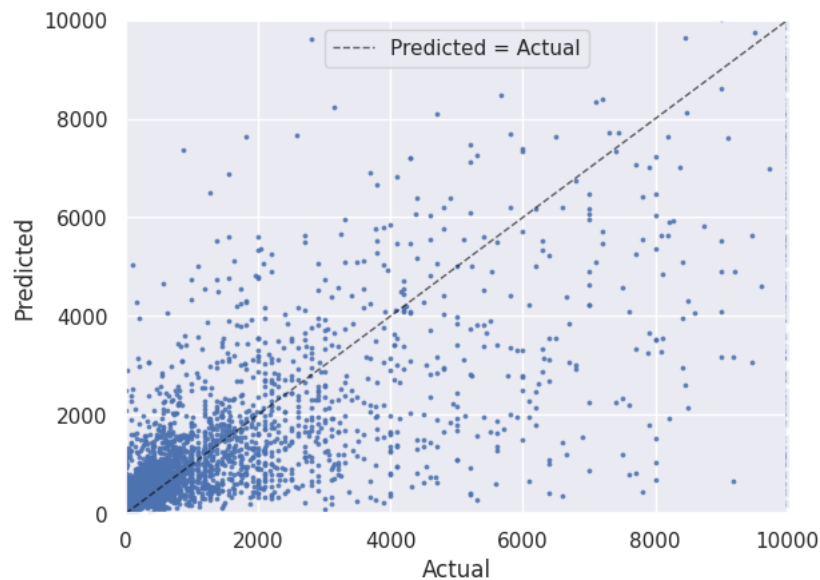
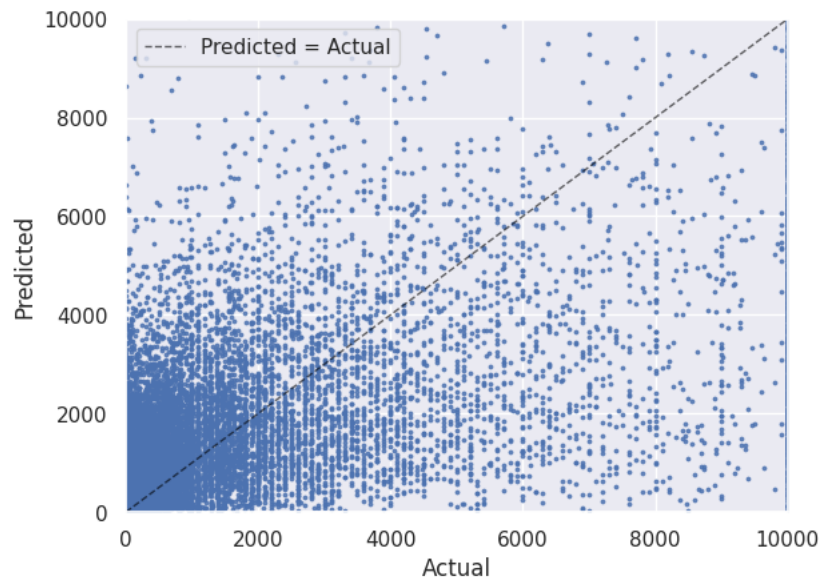


Fig. 12. random 100 samples from sampling locations for actual and model prediction on the test temporal split (a) advanced dataset (b) light dataset



(a) Advanced dataset: predicted versus actual (in CFU/100 ml)

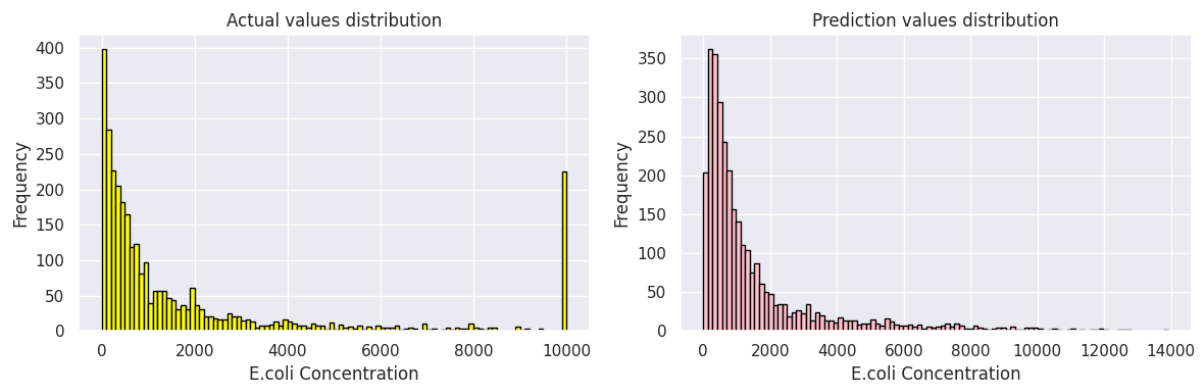


(b) Light dataset: predicted versus actual (in CFU/100 ml)

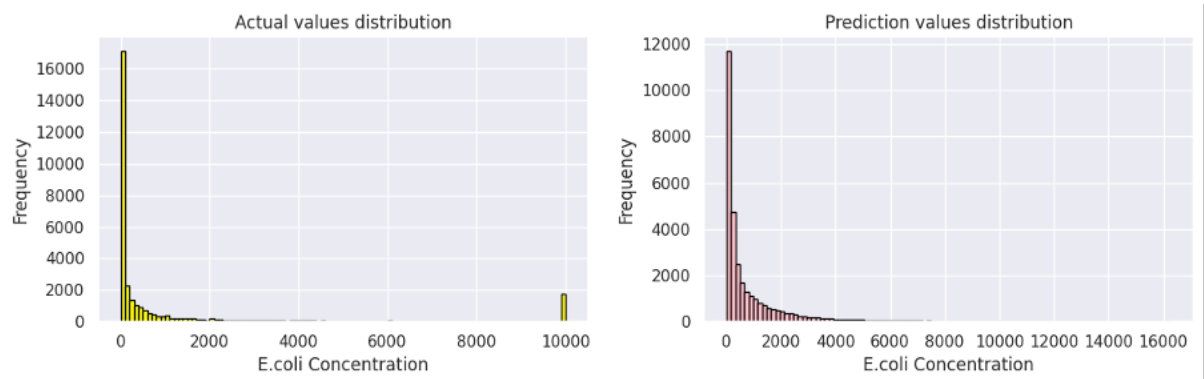
Fig. 13: scatter plot on actual and model prediction on the test temporal split. (a) advanced dataset (b) light dataset

Both plots on Fig. 13 show a high density of points concentrated in the lower range (0–2000 CFU/100 ml on both axes), indicating that most samples fall within this region. This clustering suggests that while the model to some extent performs slightly better for lower values in absolute terms, the spread increases for higher actual values, leading to larger deviations from the ideal 1:1 line. It should be noted however that the relative / percentage error is substantially larger for low measured values of *E. coli*.

In the following plots (Fig 14 and 15), we analyse the actual and prediction distribution and error margin. It should be noted that the actual values of *E. coli* were capped at 10,000, since we wanted the model to focus on lower concentration rather than very high values. However, as the data was still used for training the model, this can influence both the learning process and the predictions by the model. By limiting extreme values, the model is not influenced by extreme outliers, which may improve overall stability and reduce overfitting at the lower concentration ranges of interest. On the other hand, the model predictions can still exceed the capped value because the model estimates a continuous function of the predictors, i.e. if the input features indicate conditions similar to historically very high *E. coli* levels, the model may extrapolate beyond the capped range. This means that the model does not learn the true behaviour at *E. coli* concentrations above the capped value, making the model unreliable for extreme events. In this case, the overall metrics may be affected.

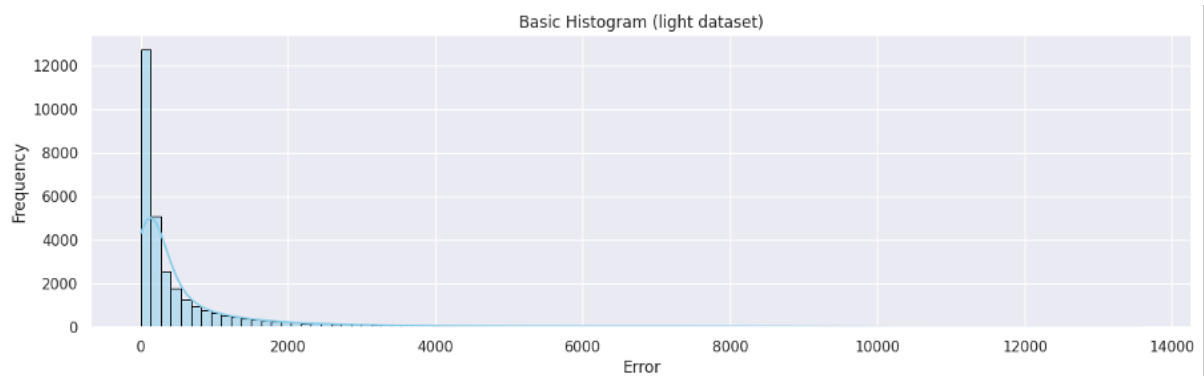
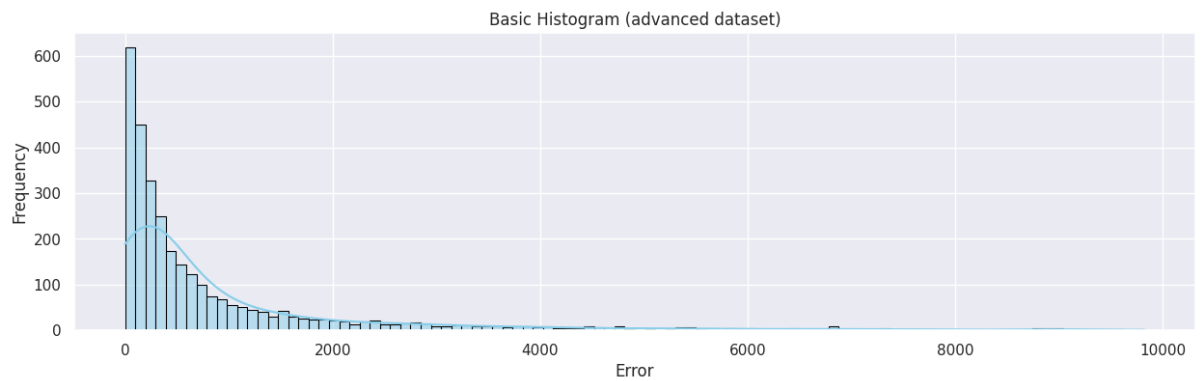


(a)

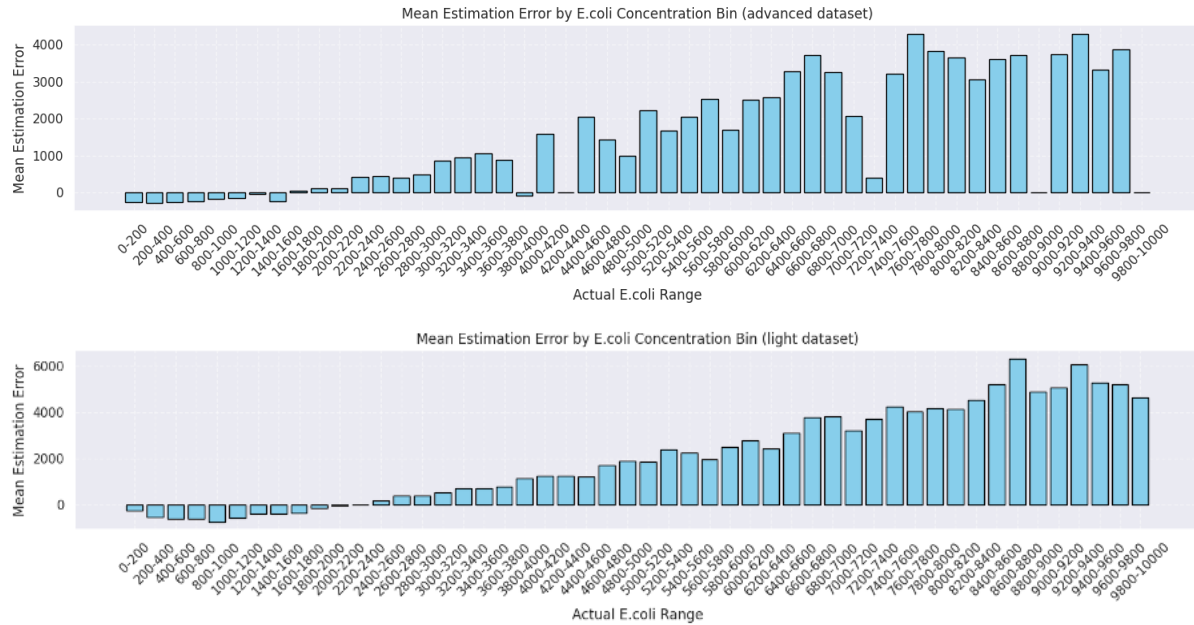


(b)

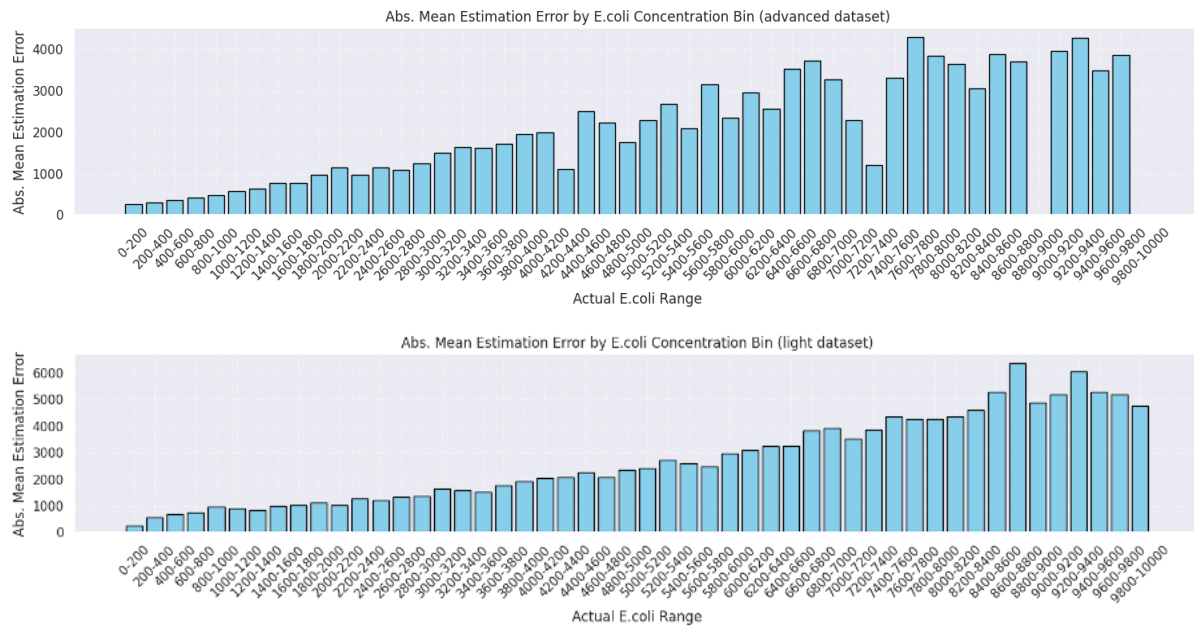
Fig. 14 the distribution of test data in histogram format. (a) advanced dataset (b) light dataset



(a) Absolute Error of model v4.0 in the histogram format (advanced and light)



(b) The mean of error/residual for Actual *E. coli* ranges/bins (advanced and light).



(c) The abs. mean of error/residual for Actual *E. coli* ranges/bins (advanced and light)

Fig.15 (a) Absolute Error of model v3.0 in the histogram format. (b) the mean of error/residual for Actual *E. coli* ranges/bins. (c) the abs. mean of error/residual for Actual *E. coli* ranges/bins.

5.2.2 Evaluating the estimation model's ability to classify risk

The graphs above detail how the *E. coli* estimator performs when estimating the specific *E. coli* concentration in CFU/100 ml. As showcased in the previous section, even though the estimator has some value in estimating *E. coli* concentrations, its

error margin expands when attempting to estimate high concentrations. Therefore, additional investigations were carried out to evaluate the model's ability to classify the risk of *E. coli* contamination. We did this with two sets of classes, based on regulatory guidelines (shown in Table IV):

- Figure 17: Three classes (low if $x < 235$ CFU/100 ml, moderate if $1000 \text{ CFU}/100 \text{ ml} \geq x \geq 235 \text{ CFU}/100 \text{ ml}$ and else high)
- Figure 18: Two classes (above/below 1000 CFU/100 ml)

To evaluate the ability of the regression model to classify risk, we have created the classification curves (Fig. 16) and confusion matrices focused on three (Fig. 17) and two classes (Fig. 18).

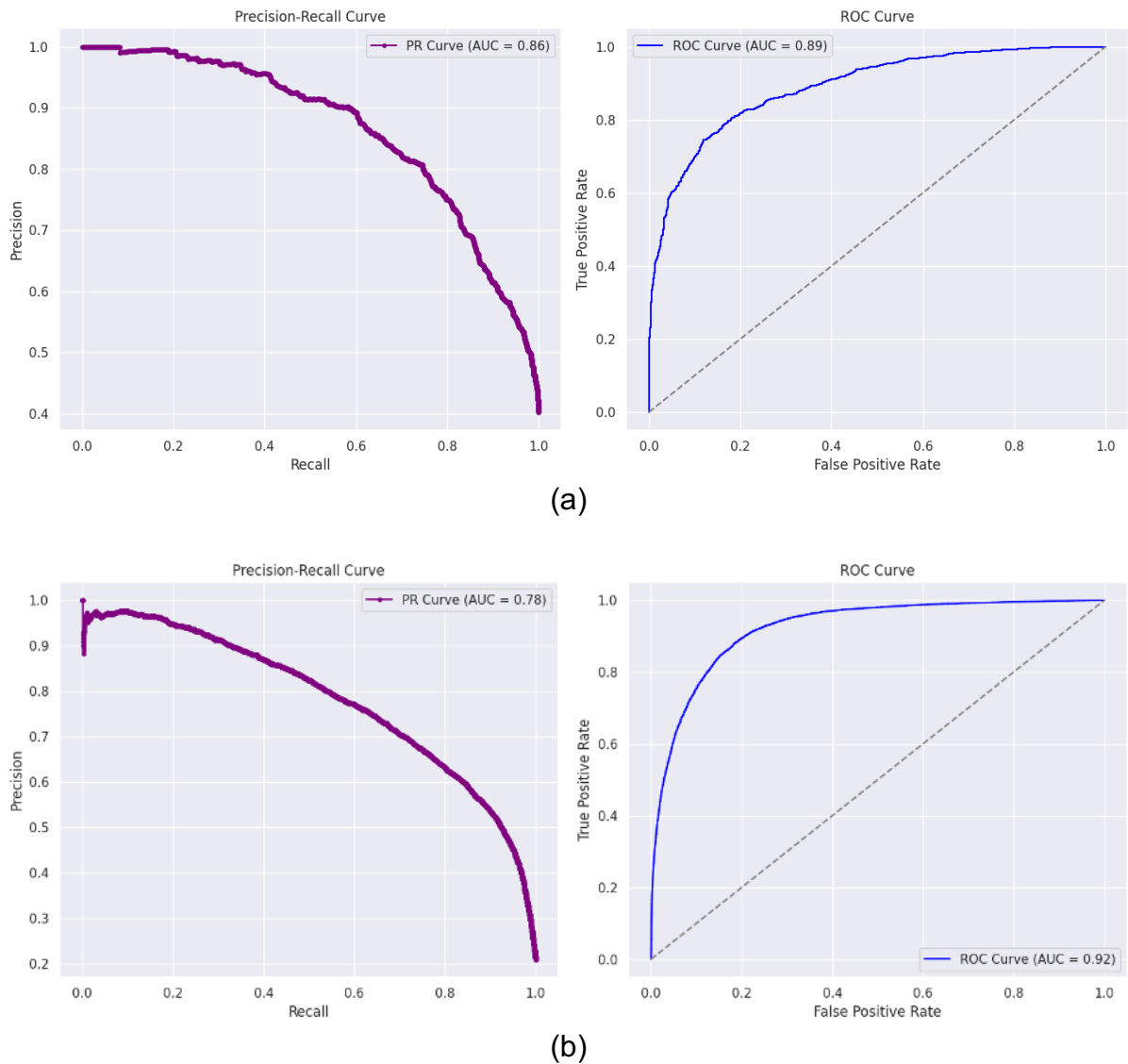
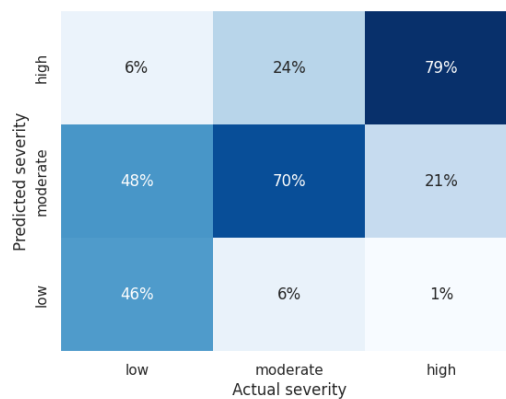


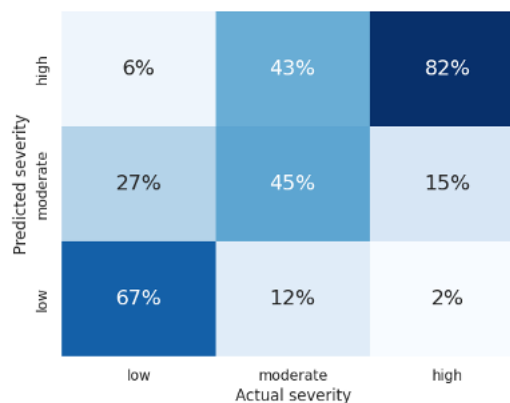
Fig. 16 Precision-Recall and ROC curves. (a) advanced dataset (b) light dataset
The continuous predictions (*E. coli* CFU/100 ml) by the regression model were treated as scores. By sweeping thresholds, PR (Precision–Recall) and ROC

(Receiver Operating Characteristic) curves were generated. The PR/POC curves were calculated in the following way: Binary classification is made on the defined threshold (i.e. either 500 CFU/100 ml or 1000 CFU/100 ml). Later predictions (i.e. model outputs) are considered as Pseudo-probabilities (which is non-thresholded measure of decisions). Then continuous threshold will apply to measure the curves [see 22, 23].

At low recall values (near 0), the precision is high (close to 1), indicating the model is being very selective but accurate when it does make positive predictions. As recall increases, precision gradually decreases, showing that the model is capturing more positives but at the cost of introducing more false positives. A good model will have a curve that stays close to the top-right corner, which means both precision and recall are high across most threshold settings. The area under the PR curve can also be an indicator of performance. The larger the area, the better the model performs in terms of precision and recall trade-offs. According to Fig. 18, we can see that both models perform well in detecting *E. coli* concentrations above the specified threshold (84% for advanced model) and detecting 78% (for advanced model) of the concentrations below the threshold.

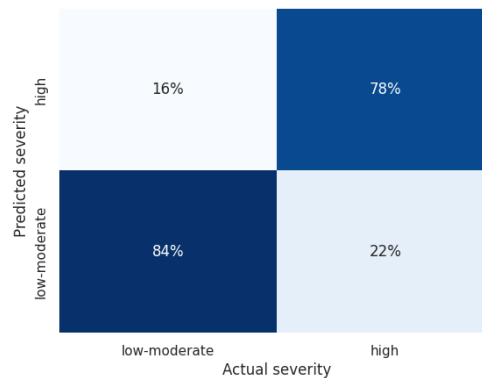


(a)

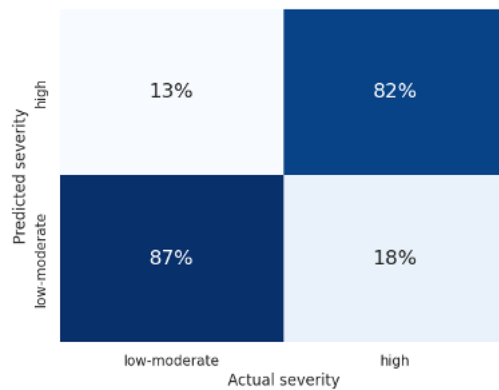


(b)

Fig. 17 Three classes confusion matrix (low if $x < 235$, moderate if $1000 \geq x \geq 235$, and else high). (a) advanced dataset (b) light dataset



(a)



(b)

Fig 18 Two classes confusion matrix (threshold: 1000 CFU/100 ml). (a) advanced dataset (b) light dataset

For further detail on the above, please see the JupyterLab output .pdf file or .ipynb notebook for the models available in GitHub[21].

5.3 Classifier Performance Analysis

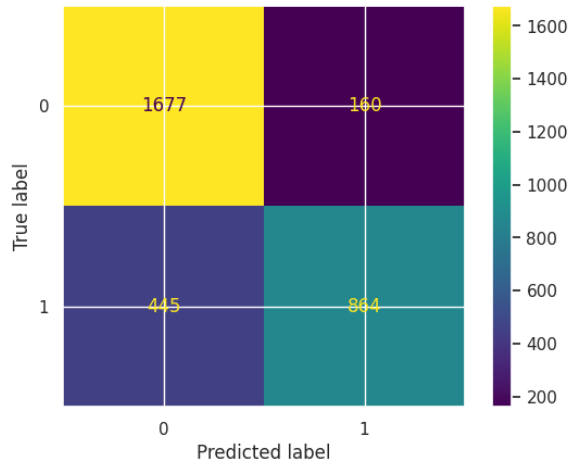
In this section we detail the performance metrics for the classifier model, capable of classifying the risk of *E. coli* contamination. As with the other models, and as described in section 3.1, the classifier models are based on a 500 CFU/100 ml and 1000 CFU/100 ml threshold.

The following will detail the performance results for our **XGBClassifier** model from the temporal split training as this showed the best performance across metrics (see section 5.1). Below we can see per class performance on various metrics.

Table VI: Performance of XGBClassifier trained over the temporal split

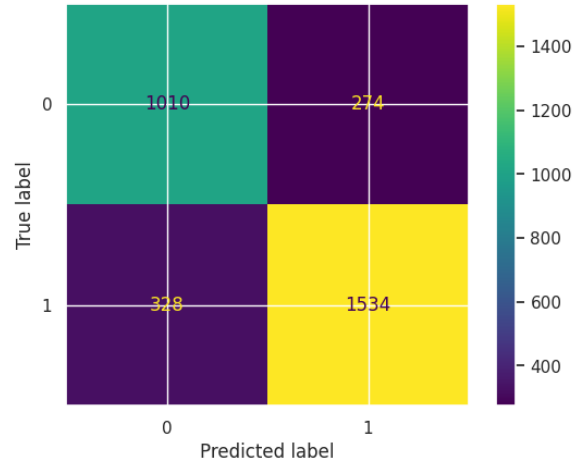
Metrics	Threshold: 1000 CFU/100 ml	Threshold: 500 CFU/100 ml
---------	----------------------------	---------------------------

	Advanced	Light	Advanced	Light
Accuracy	0.8077	0.8761	0.8086	0.8674
F1 Score	0.8029	0.8668	0.8092	0.8637
Recall	0.8077	0.8761	0.8086	0.8674
Precision	0.8125	0.8713	0.8103	0.8649
False Positives (FP)	0.0575	0.0302	0.0943	0.0459
False Negatives (FN)	0.1371	0.0951	0.1023	0.0889

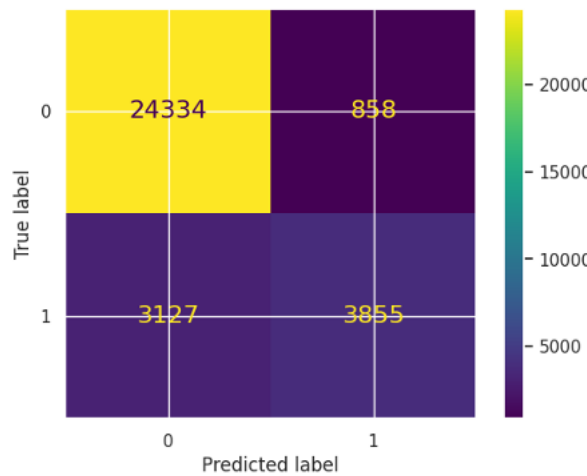


(a) Advanced (1000 CFU/100 ml)

(500 CFU/100 ml)

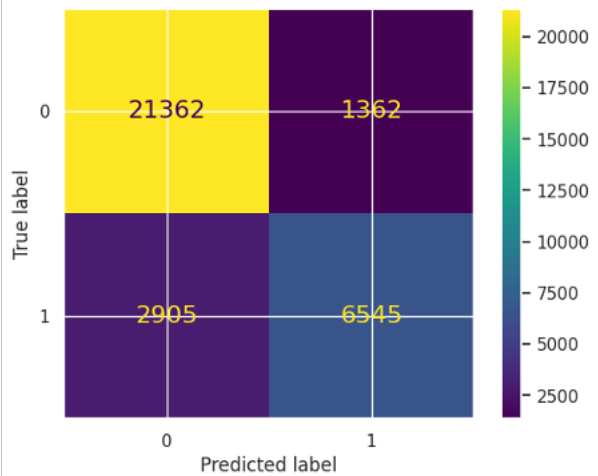


(b) Advanced



(c) Light (1000 CFU/100 ml)

CFU/100 ml)



(d) Light (500

Fig. 19 Two classes confusion matrix for temporal split. 0 is below threshold and 1 is above threshold. (a) on advanced dataset for threshold: 1000 CFU/100 ml (b) on advanced dataset for threshold: 500 CFU/100 ml (c) on light dataset for threshold: 1000 CFU/100 ml (d) on light dataset for threshold: 500 CFU/100 ml

Different from the estimator model which has somewhat balanced performance on both classes, the classifier model performs better when classifying low risks of *E. coli* concentration levels (especially on the light dataset) with respect to Fig. 19. This can be due to the imbalance in the data as we have lower number of high *E. coli* samples

in the advanced set. In Fig. 20, The PR curves indicate that the model maintains relatively high precision even as recall increases — particularly in the light dataset.

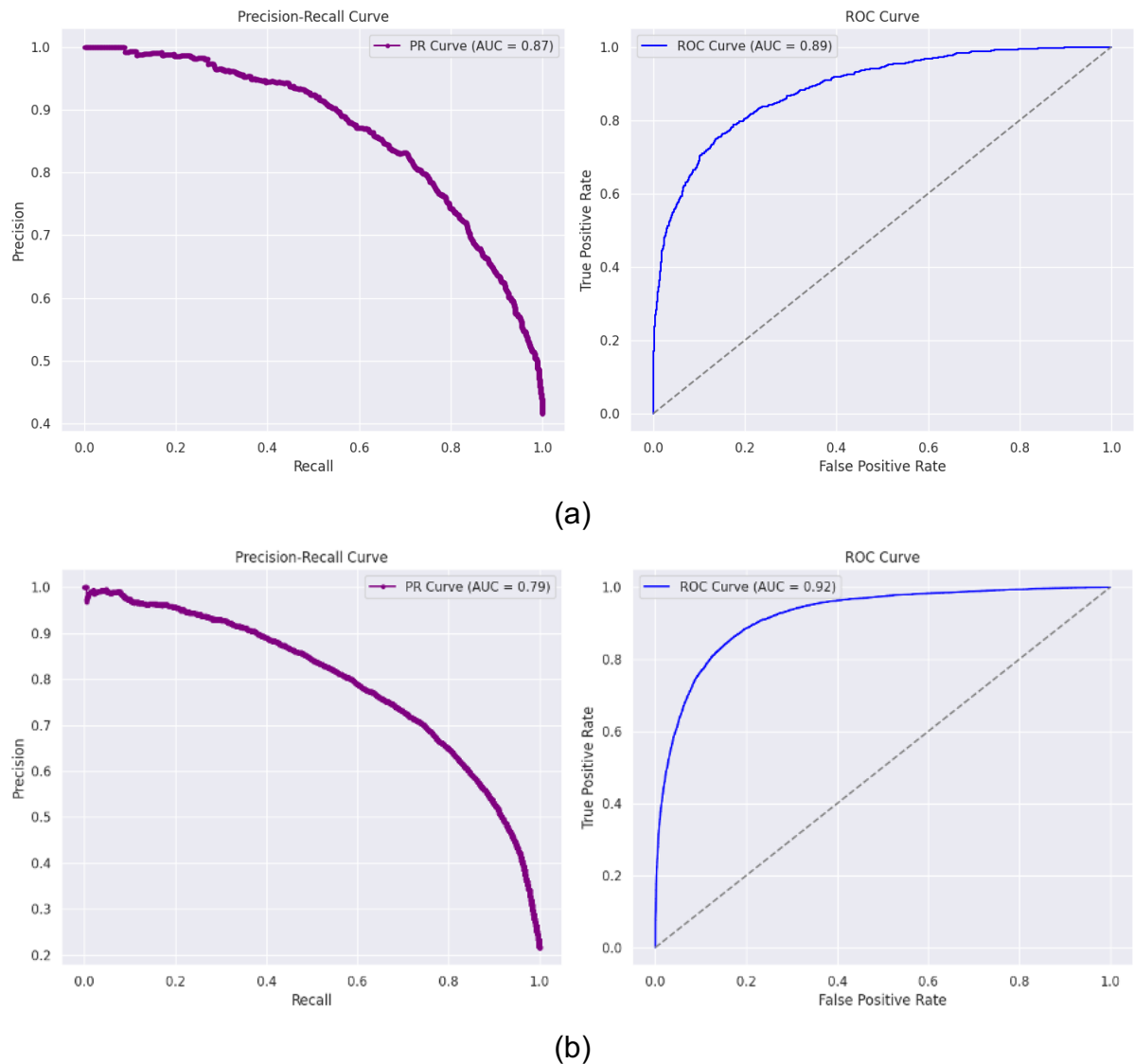
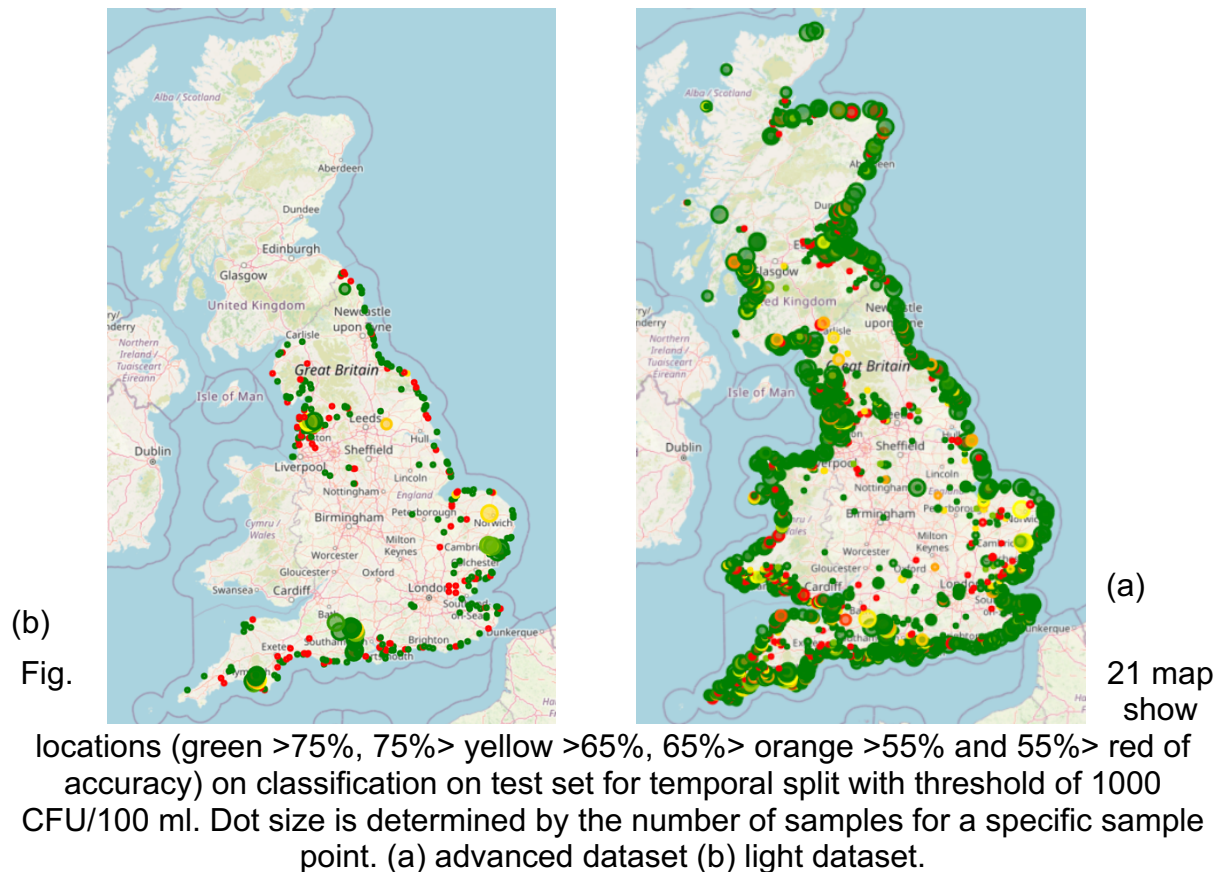


Fig. 20 Two classes Precision-Recall and ROC curves (threshold: 1000 CFU/100 ml).
(a) advanced dataset (b) light dataset

Fig. 20 (above) shows the PR and ROC curves for both advanced and light models. A higher AUC for a Precision-Recall (PR) curve and a lower AUC for a ROC curve together often indicate a model with good performance on a highly imbalanced dataset (i.e. light dataset), particularly regarding the positive minority class.

The PR curve is sensitive to class imbalance and focuses on the positive class, so a higher PR AUC suggests the model identifies true positives while minimizing false positives relative to the small number of actual positives. Conversely, the ROC curve is less sensitive to class imbalance.



5.4 General Observations

- All final version 4 models perform significantly better than earlier baselines (version 1 through 3).
- It can be observed that our light models achieve higher performance in comparison to advanced models, with this difference in performance more visible in the classifier models. Although, it is important to note that it might not be a fair comparison, as light and advanced models have been tested on different datasets.
- Training datasets for both advanced and light models are unbalanced, with a low number of high-concentration *E. coli* samples. These dataset imbalances were alleviated to some extent when the threshold was reduced to 500 CFU/100 ml*.

*We tried to augment the high concentration class by duplicating these class's samples and adding low power additive noise, but no visible improvement in performance has been seen.

- **Light model:** Light classifier model achieved **highest classification accuracy** overall (87.61%) and F1 (0.86) over temporal split. Although, it is important to note that its per-class classification performance was imbalanced and performed better in detecting lower concentration of *E. coli*. On the other

hand, the light regressor model, with accuracy of 85.61%, provided somewhat balanced performance over both classes.

- **Advanced model:** The advanced regressor model achieved a **high classification accuracy** (81.72%) over temporal split. Although over geographical split, it achieves higher performance (83.49%), but the per class performance is unbalanced.
- **Geospatial split:** We note a drop in performance in geospatial split (see section 5.1) which indicates some overfitting to locations and/or existence of catchment specific characteristics.
- **Estimation models:** Low R^2 (0.142 for the light model), indicating very poor variance explanation.

5.4.1 Key Performance Limits Identified

Geographic generalisation:

1. Both the classification and estimation models suffer from significant performance drop in geographic split (see section 5.1), especially the light models.
2. Performance comparison on temporal and geographical splits indicates overfitting to known locations — a critical limitation for models intended for real-world deployment across diverse coastal areas. This is less of an issue when the model is used in already designated bathing waters, however, when used in newly designated bathing waters (inland and coastal) the model will have limitations.
3. Higher accuracy performance on temporal split indicates models rely on localised patterns, which don't translate well to geographical split. Drop in performance over geographical split suggests strong location-dependence on the models or indicates likely overfitting over temporal patterns on temporal split.

Light vs. Advanced Trade-off:

1. Light models (without in-situ water quality data) achieve better performance in temporal split (better suited for real-time, low-resource scenarios). In essence, the light model is designed to be used in locations where no in-situ water quality parameter is available. On the other hand, advanced models are more solid in geographical split, suggesting in-situ water quality data is key for generalisation.

False Negatives (FN):

1. FN and FP are considerably higher in geographical split (especially for the light models), indicating generalisation issues (see section 5.1).

2. Higher FN, especially on light models, raises concerns about under-detection of high-risk events, which are critical for public health.

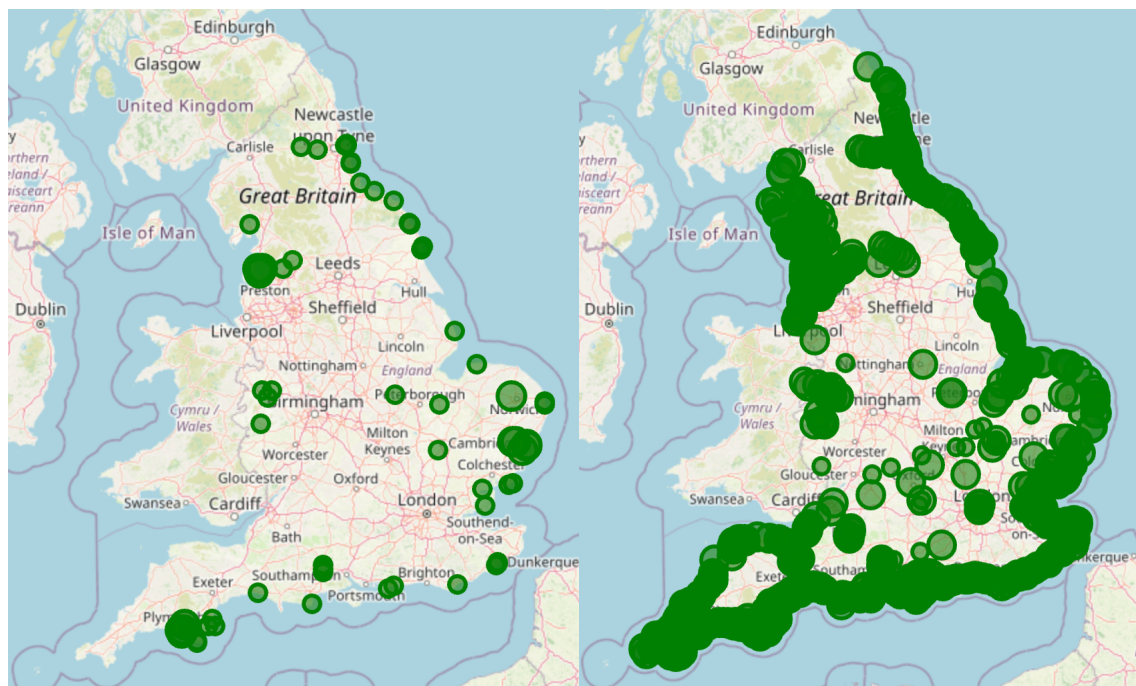
FN is consistently higher than FP in most setups — a negative trend, as missing contaminated sites has greater public health implications.

6. Validation

In this section we will go through the performance validation of our two approaches, estimation and classification (advanced). Initially we provide an overview of validation set for both advanced and light models. Later, in the following sections, we detail the performance of the individual models.

6.1 Validation Set: Environment Agency – Water Quality Archive (WIMS)

To validate our models, we have used *E. coli* measurements from the Environment Agency sampled from July 2024 to July 2025. The EA dataset contains about **12,000** *E. coli* samples from **1,029** locations in the England, which we have used to validate our light model. Out of these, filtering based on select proxy parameters (detailed in section 2.1.1) reduces this to **649 usable samples** based on the advanced model requirements. These 649 samples have been used as our advanced model validation set.



(a) Advanced catchments (b) Light catchments

Fig. 22. Catchment locations for validation sets. Dot size is determined by the number of samples for a specific sample point. (a) advanced (b) light

6.2 Overview of performance

In this section we outline the validation performance metrics of the best performing models (XGBClassifier and XGBRegressor). For full detail on the performance of the

other algorithms, please refer to the JupyterLab output pdf file available in GitHub [21].

We will outline the model performance split between advanced and light versions, 500 CFU/100 ml and 1000 CFU/100 ml and compare train/test performance versus validation performance. Please note, for the reporting of FP and FN in Table VII below, the number outside the bracket is the proportion among all categories (FP, FN, TP and TN), whereas the number in the bracket for FN is the proportion among FN and TP (positive class) and the number in the bracket for FP is proportion among FP and TN (negative class).

Table VII: The validation performance of version 4 for *E. coli* classification (top table) and estimation (bottom table).
E. coli Classifier Random Temporal Split Training Random Geographic Split Training

<i>Model _dataset_thr</i>	Accuracy	Precision	Recall	F1-score	FP (Per class)	FN (Per class)	Accuracy	Precision	Recall	F1-score	FP (Per class)	FN (Per class)
Advanced train/test metrics:												
<i>XGBCv4 adv 1000</i>	80.77	0.81	0.80	0.80	0.057	0.137	75.29	0.80	0.75	0.76	0.052 (0.17)	0.194 (0.28)
<i>XGBCv4 adv 500</i>	80.86	0.80	0.80	0.80	0.094	0.102	80.36	0.82	0.80	0.81	0.073 (0.40)	0.123 (0.15)
Advanced validation metrics:												
<i>Validation: XGBCv4 adv 1000</i>	78.43	0.78	0.78	0.78	0.083	0.132	77.35	0.77	0.77	0.76	0.071 (0.10)	0.156 (0.44)
<i>Validation: XGBCv4 adv 500</i>	80.43	0.80	0.80	0.80	0.111	0.085	78.74	0.79	0.79	0.79	0.095	0.117
Light train/test metrics:												
<i>XGBCv4 lgt 1000</i>	87.61	0.87	0.87	0.86	0.030 (0.04)	0.095 (0.44)	80.07	0.77	0.80	0.77	0.049 (0.06)	0.145 (0.70)
<i>XGBCv4 lgt 500</i>	86.74	0.86	0.86	0.86	0.046 (0.06)	0.089 (0.30)	73.81	0.72	0.74	0.73	0.092 (0.14)	0.181 (0.56)
Light validation metrics:												
<i>Validation: XGBCv4 lgt 1000</i>	80.97	0.78	0.81	0.74	0.007 (0.01)	0.182 (0.91)	82.83	0.81	0.83	0.79	0.020 (0.02)	0.152 (0.76)
<i>Validation: XGBCv4 lgt 500</i>	75.15	0.74	0.75	0.70	0.023 (0.01)	0.224 (0.81)	79.70	0.79	0.80	0.78	0.039 (0.05)	0.164 (0.59)

Top table: *E. coli* classifier performance split between advanced and light versions.

E. coli Estimator

Random Temporal Split Training

Random Geographic Split Training

<i>Model</i>	Accuracy	MAE	R2	RMSE	FP (Per class)	FN (Per class)	Accuracy	MAE	RMSE	R2	FP (Per class)	FN (Per class)
Advanced train/test metrics:												
<i>XGBRv4 adv 1000</i>	81.72	984	0.557	1870	0.090	0.092	77.50	1883	2935	0.282	0.085	0.139
<i>XGBRv4 adv 500</i>	80.51	984	0.557	1870	0.127	0.067	83.49	1883	2935	0.282	0.099 (0.52)	0.056 (0.08)
Advanced validation metrics:												
<i>Validation: XGBRv4 adv 1000</i>	72.11	1213	0.023	2450	0.035 (0.05)	0.243 (0.70)	72.72	1212	2499	0.023	0.029 (0.05)	0.243 (0.70)
<i>Validation: XGBRv4 adv 500</i>	69.18	1213	0.023	2450	0.067 (0.16)	0.24 (0.42)	69.33	1212	2499	0.023	0.067	0.238
Light train/test metrics:												
<i>XGBRv4 lgt 1000</i>	85.61	845	0.497	1808	0.103	0.040	74.13	1227	2349	0.142	0.183 (0.22)	0.095 (0.41)
<i>XGBRv4 lgt 500</i>	82.48	845	0.497	1808	0.145	0.030	65.76	1227	2349	0.142	0.294 (0.39)	0.075 (0.24)
Light validation metrics:												
<i>Validation: XGBRv4 lgt 1000</i>	83.41	941	0.257	2092	0.076 (0.10)	0.089 (0.45)	82.82	961	2198	0.180	0.062 (0.08)	0.109 (0.55)
<i>Validation: XGBRv4 lgt 500</i>	77.83	941	0.257	2092	0.147 (0.20)	0.074 (0.27)	77.23	961	2198	0.180	0.127 (0.18)	0.100 (0.36)

Bottom table: *E. coli* estimator performance split between advanced and light versions.

Based on the results presented in Table VII (top and bottom), the XGBClassifier advanced model trained with a temporal split can be considered the best performing model, with an overall accuracy of 80.43% on the validation set. Therefore, in the following sections, we provide a more detailed examination of the validation results and related performance metrics for this particular model. For further validation analysis of *E. coli* model under the light configuration, we focus on the XGBRegressor light model that was trained with a temporal split.

It is important to mention that since the **XGBClassifier** is unable to provide concentration estimations, either of the **XGBRegressors** (500 and 1000 threshold) shown in Table VII can be used if concentration estimations are required.

The following sections will go into details with the performance metrics of the **XGBRegressor Light** and **XGBClassifier Advanced** (both with 500 CFU/100 ml threshold).

6.2.1 Regression metrics

Fig. 24 (below) represents the model performance for the XGBRegressor light estimator model trained on a temporal split (detailed in section 5).

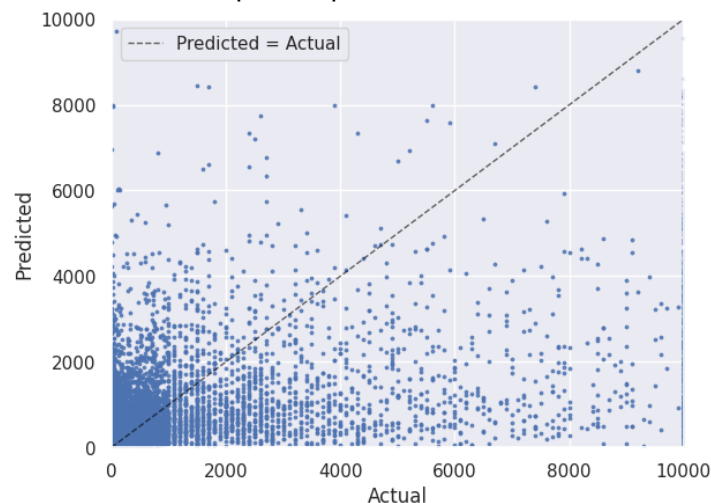


Fig. 24: scatter plot of actual and XGBRegressor model prediction on the light validation set.

The following plots (Fig 25 and 26) provide the actual and prediction distribution and error margin of the light model. It is worth noting as in the training process, we capped the actual values at 10,000, while the output of the model remained unrestricted.

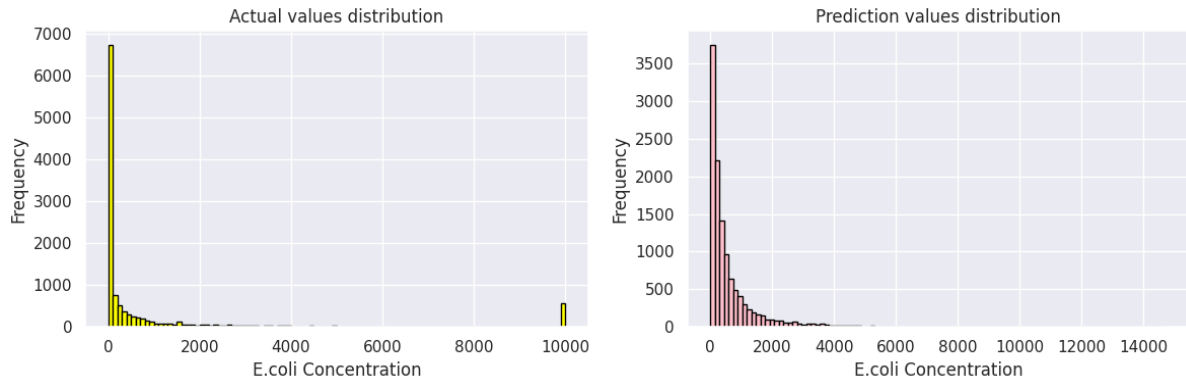
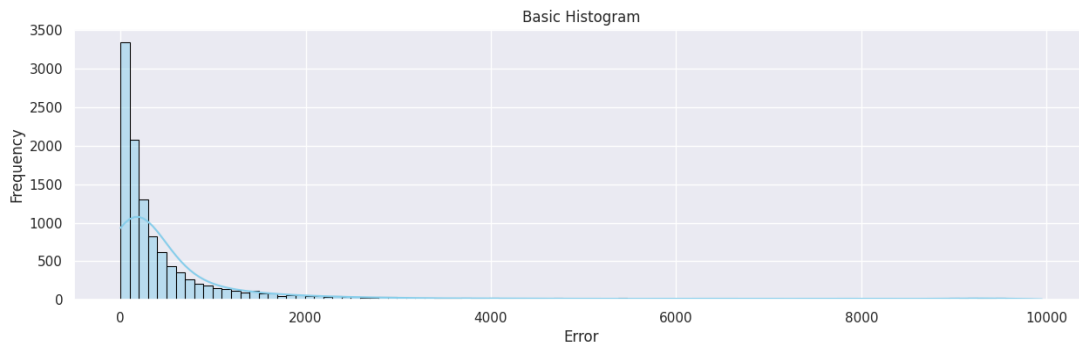
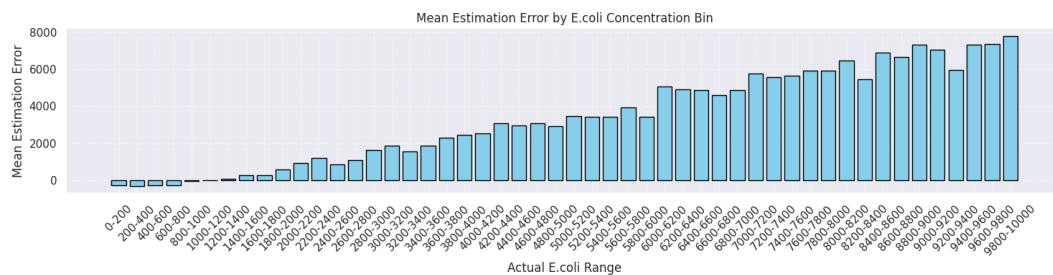


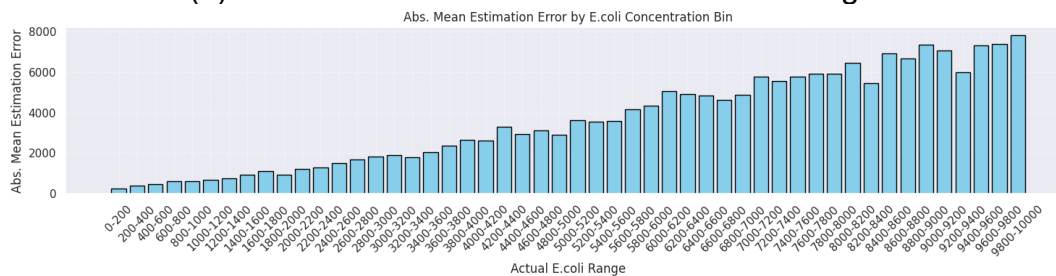
Fig. 25 the distribution of light validation set in histogram format.



(a) Absolute Error of XGBRegressor light model v4.0 on the light validation set.



(b) Mean of error/residual for actual *E. coli* ranges



(c) MAE for actual *E. coli* ranges

Fig. 26 (a) Absolute Error of XGBRegressor light model v4.0 on the light validation set in the histogram format . (b) the mean of error/residual for Actual *E. coli* ranges/bins. (c) the abs. mean of error/residual for Actual *E. coli* ranges/bins. Figure 26 above details how the XGBRegressor light estimator performs when estimating the specific *E. coli* concentration in CFU/100ml. Similar to previous analysis on section 5.2.1, even though the estimator has some value in estimating *E. coli* concentrations, its error margin expands when attempting to estimate high concentrations. There is a possibility that the overall errors could be artificially inflated because the predictions were not also capped at 10,000. Therefore, the

errors relating to values >10,000 could make the model appear worse than it actually is. This is something worth exploring in further work.

6.2.2 Classification metrics

The following section will go into details with the performance metrics of the **XGBRegressor Light** and **XGBClassifier Advanced** (both with 500 CFU/100 ml threshold) on classification metrics. We look into both, as they represent the advanced and light version of the model with the highest performance in the validation process. Both can be used for classifying *E. coli*. The following explores their performance.

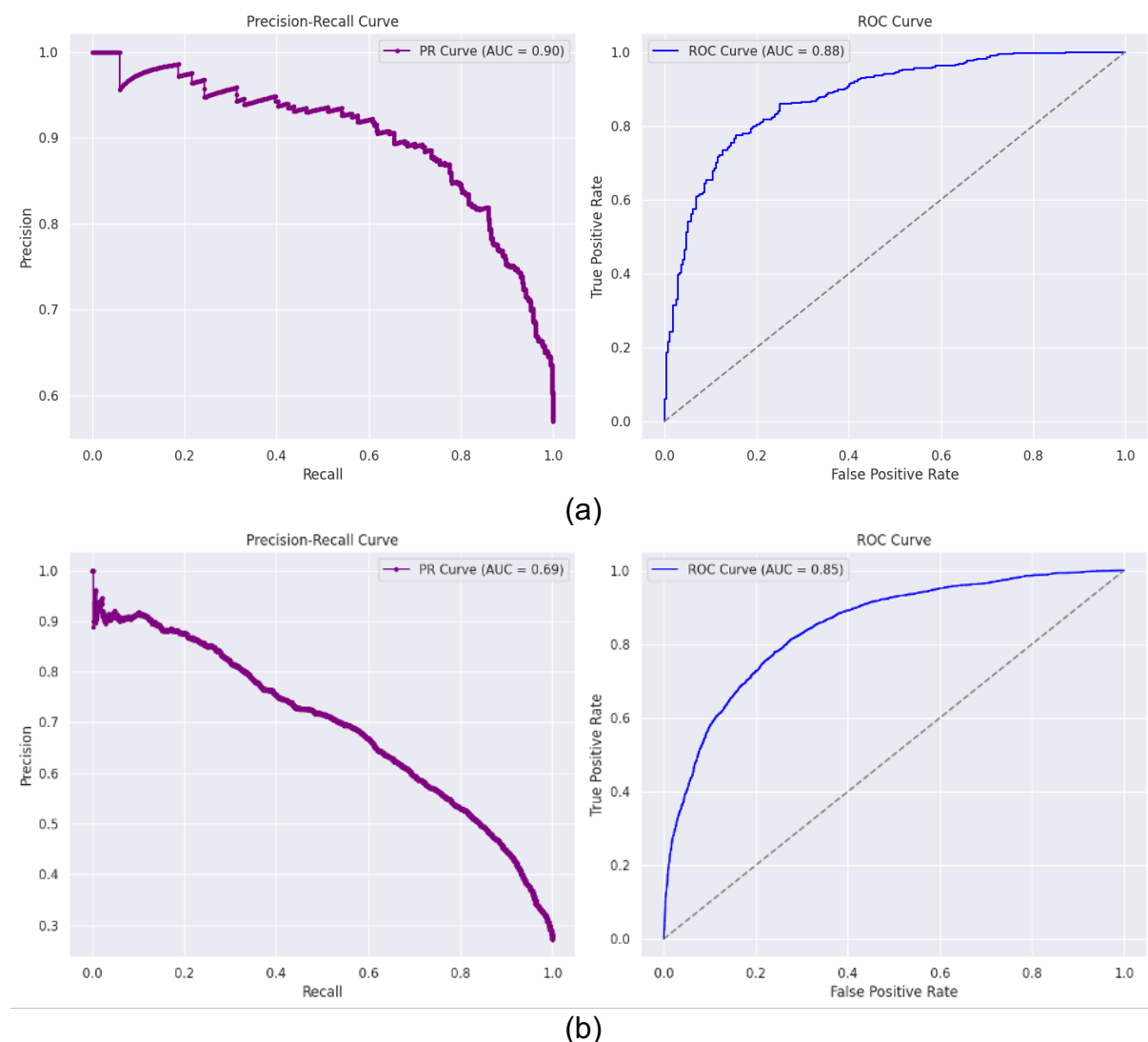


Fig. 27 Precision-Recall and ROC curves (threshold: 500 CFU100 ml). (a) advanced XGBClassifier (b) light XGBRegressor

Fig. 27 shows the PR and ROC curves for both advanced and light model. The PR AUC quantifies how well a model can distinguish between classes. Since the advanced XGBClassifier has higher PR AUC, this is considered the better-

performing model. However, it should be noted that the validation sets are different: the advanced XGBClassifier performance is provided with respect to the advanced validation set, while the other model uses the light validation set.

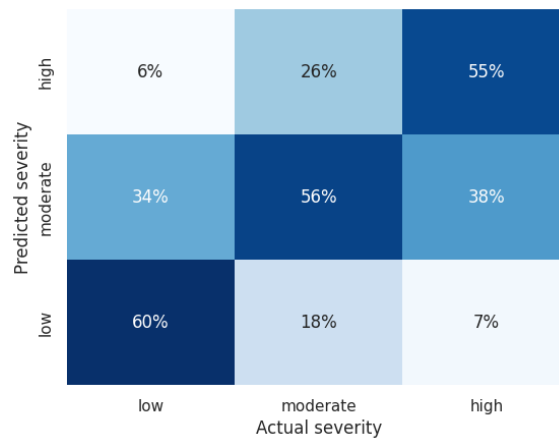
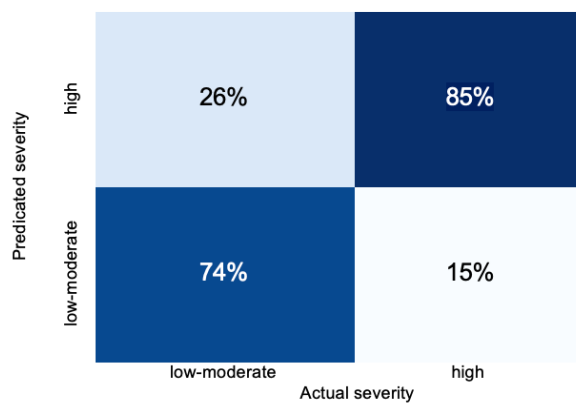
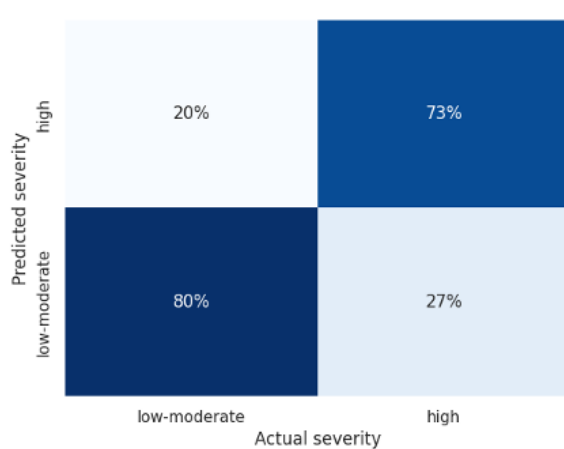


Fig. 28 Three classes confusion matrix (low if $x < 235$, moderate if $1000 \geq x \geq 235$, and else high) for the light XGBRegressor on validation set



(a)



(b)

Fig 29 Two classes confusion matrix on validation set (threshold: 500 CFU/100 ml).
(a) advanced XGBClassifier (b) light XGBRegressor

Fig. 28 and 29 illustrates the light XGBRegressor has better true negative percentage (i.e. 80% vs 74%). Conversely, the advanced XGBClassifier shows better high concentration *E. coli* detection (i.e. 85% vs 73% on true positive). Moreover, both models have somewhat balanced performance on both classes.

6.3 Key Observations (validation)

- To understand the usefulness of these models for bathing water quality assessment, the key parameters to compare are the true positives (TPs) and false negatives (FNs), which we would want to maximise and minimise, respectively, in the interest of precautionary assessment of *E. coli* concentration. The FN percentages are 15% for the advanced XGBClassifier and 27% for the light XGBRegressor using the 500 CFU/100 ml threshold (Fig. 29). Therefore, the advanced model is more likely than the light model to predict when actual high *E. coli* concentrations are present. However, neither model performs sufficiently to be used on their own for bathing water quality assessment at this stage; rather, *E. coli* concentration should be estimated in conjunction with other methods. For the purposes of bathing water quality assessment, or other applications such as environmental modelling or scenario analysis, the models should be used as an indication of *E. coli* concentrations only. The models are not appropriate for the prediction of human health risk.
- According to table VII, although accuracy is high, imbalanced performance is observed on the validation of most models, which suggests that most of models fail to generalise to unseen EA data — possibly due to many possibilities, including:
 - overfitting
 - data imbalance
 - important features are missing
 - natural variability in *E. coli* is too high to predict reliably at fine granularity
 - or significant distribution shift in features.
- Best performing advanced model is **XGBclassifier** trained over temporal split with accuracy up to 80.43% (for threshold 500 CFU/100 ml). This model was chosen due to its balanced performance.
- Best performing light model is XGBRegressor trained over temporal split with accuracy up to **83.4%** (for threshold 1000 CFU/100ml). The model performance is more reliable when 500 cfu/100ml is chosen as threshold, even though its accuracy declines slightly to 77.83% due to balanced performance per-class.
- When water quality parameters (see section 2.1) are available, we recommend prioritising the advanced **XGBClassifier** (either with the 500 or 1000 threshold) for practical deployment. When *in-situ* water quality data is not available, light XGBRegressor should be prioritised.

This work highlights the importance of coincident and colocated measurement of different water quality parameters for use in the advanced model. Much of the available EA data could not be used, despite the relevant parameters having had been measured at a particular EA monitoring point. This is because these parameters were measured on different days, thereby preventing the use of a full row of features for a particular date for the advanced model. Greater coordination of sampling programs to adjust measurement schedules would result in a significantly more useful dataset with which to carry out future similar work.

6.3.3 Key Limitations

1. **Model Bias Toward Majority Class:**
 - Except for advanced **XGBClassifier**, models show relatively high False negatives (FN): This often indicates that the model favors the majority ("safe") class.
2. **Threshold Sensitivity:**
 - Changing thresholds significantly affects FN/FP balance.

7 Future Improvement Recommendations

1. As mentioned in section 3.2, it would be beneficial to re-run the train/test split and re-train the models a few times to estimate the uncertainty of the performance metrics calculated for the models.
2. To reduce false negatives (FN) we recommend:
 - Class balancing (resampling techniques).
 - Adjusting decision thresholds.
 - Cost-sensitive training might be helpful.
3. For land cover and weather data: It might be beneficial to investigate the improvements that can be achieved by improving the upstream catchment area used to create weather and land cover input data.
4. Investigate ensemble strategies (e.g., regression + classification pipeline).
5. Add uncertainty estimation or confidence intervals — especially important due to natural noise in environmental data.
6. There is a possibility that the overall errors of the estimator models could be artificially inflated because the predictions were not capped at 10,000 (whereas the input data was capped). Therefore, the errors relating to values >10,000 could make the model appear worse than it actually is. This is something worth exploring in further work.

For any further questions or support, please refer to the documentation or reach out to hossein.darvishi@cognizant.com and nicolai.tarp@cognizant.com.

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Appendix A:

List of Parameters and cutoff points in outlier removal process on advanced dataset:

SId Sus@105C
 1950.0
 Ammonia(N)
 71.7
 Nitrite-N
 1.1
 Phosphate
 31.0
 E.coli C-MF

1200000.0
precipitation_11am
5.5
snowfall_11am
0.28
snowfall_sum
6.0199995
snowfall_sum7
16.939999
others
7.526269650376289

Appendix B:

List of Parameters and cutoff points in outlier removal process on light dataset:

Ecoli
2880000.0
others
19.76381227932483
precipitation_11am
6.0
rain_11am
5.6
snowfall_11am
0.28
snowfall_sum
2.5199997
snowfall_sum7
8.4